Anatomy of the Cervical Spine

PT5000

Tuesday, 17th February 2026

Dr Michelle Lussier

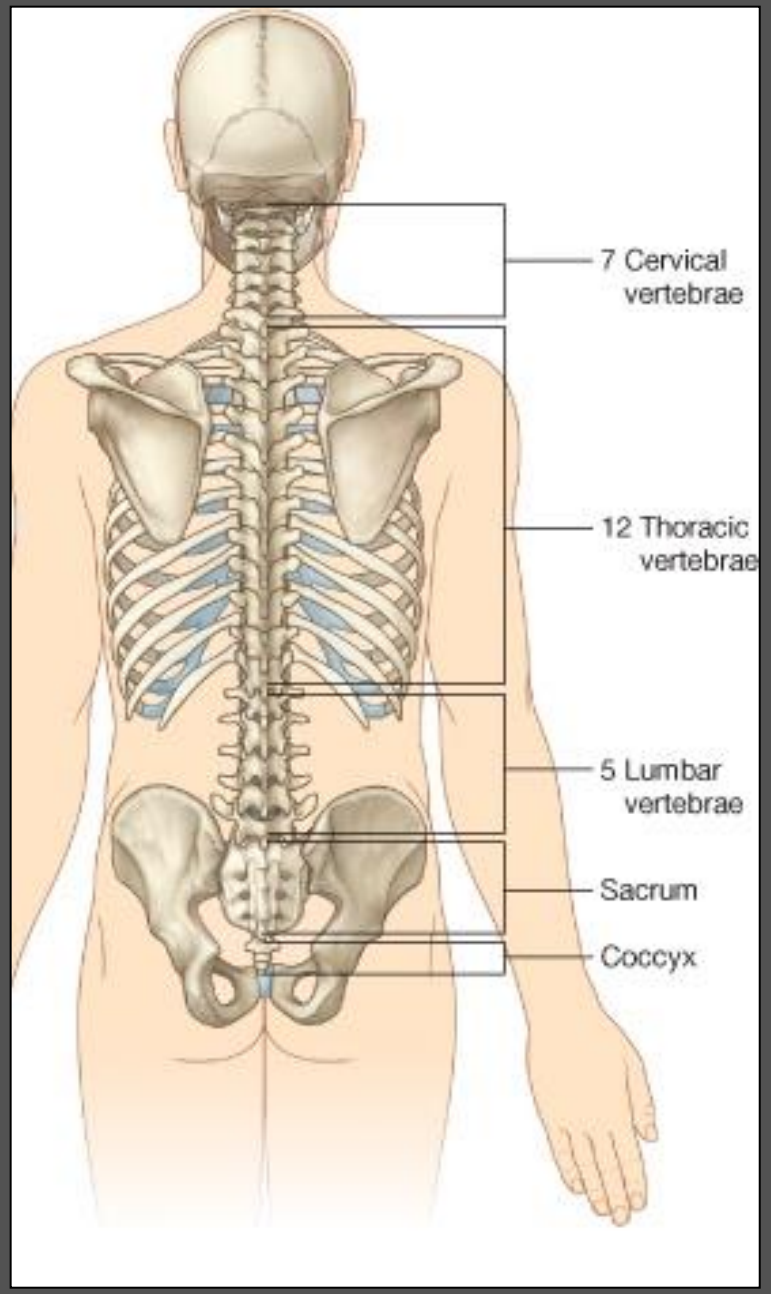
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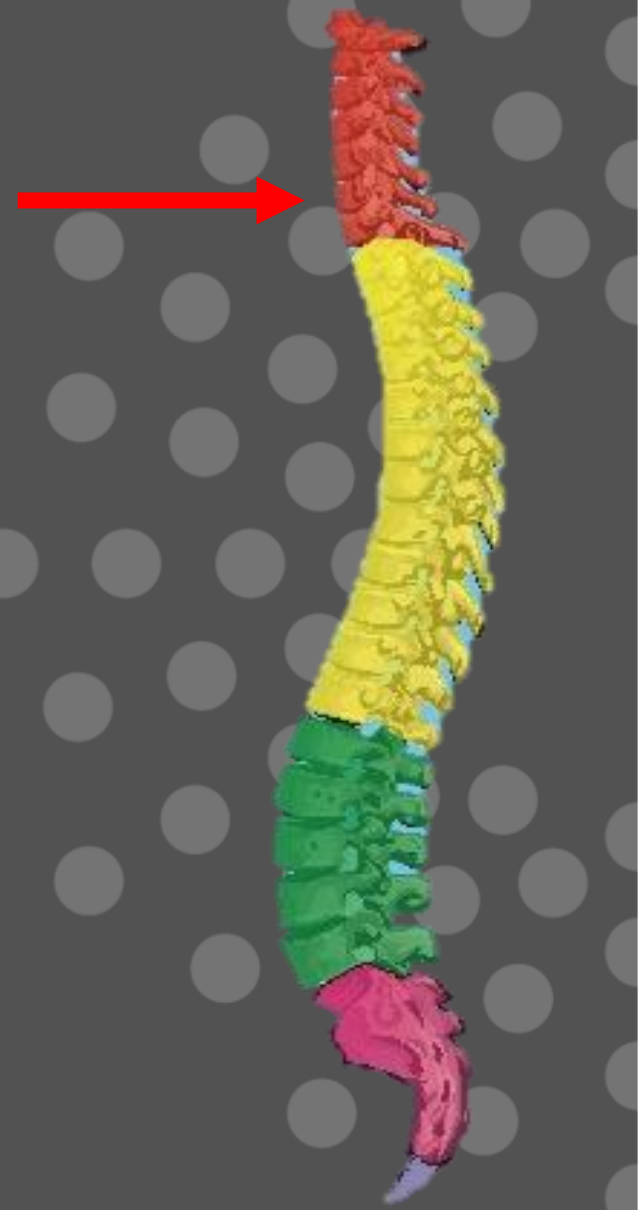
Contents

- Characteristic of Cervical vertebrae
- Atlas (C1) and Axis (C2) and their articulations
- Differences between Cervical, Thoracic and Lumbar regions
- Nerve roots of Cervical spine
- Vertebrobasilar system

Cervical spine

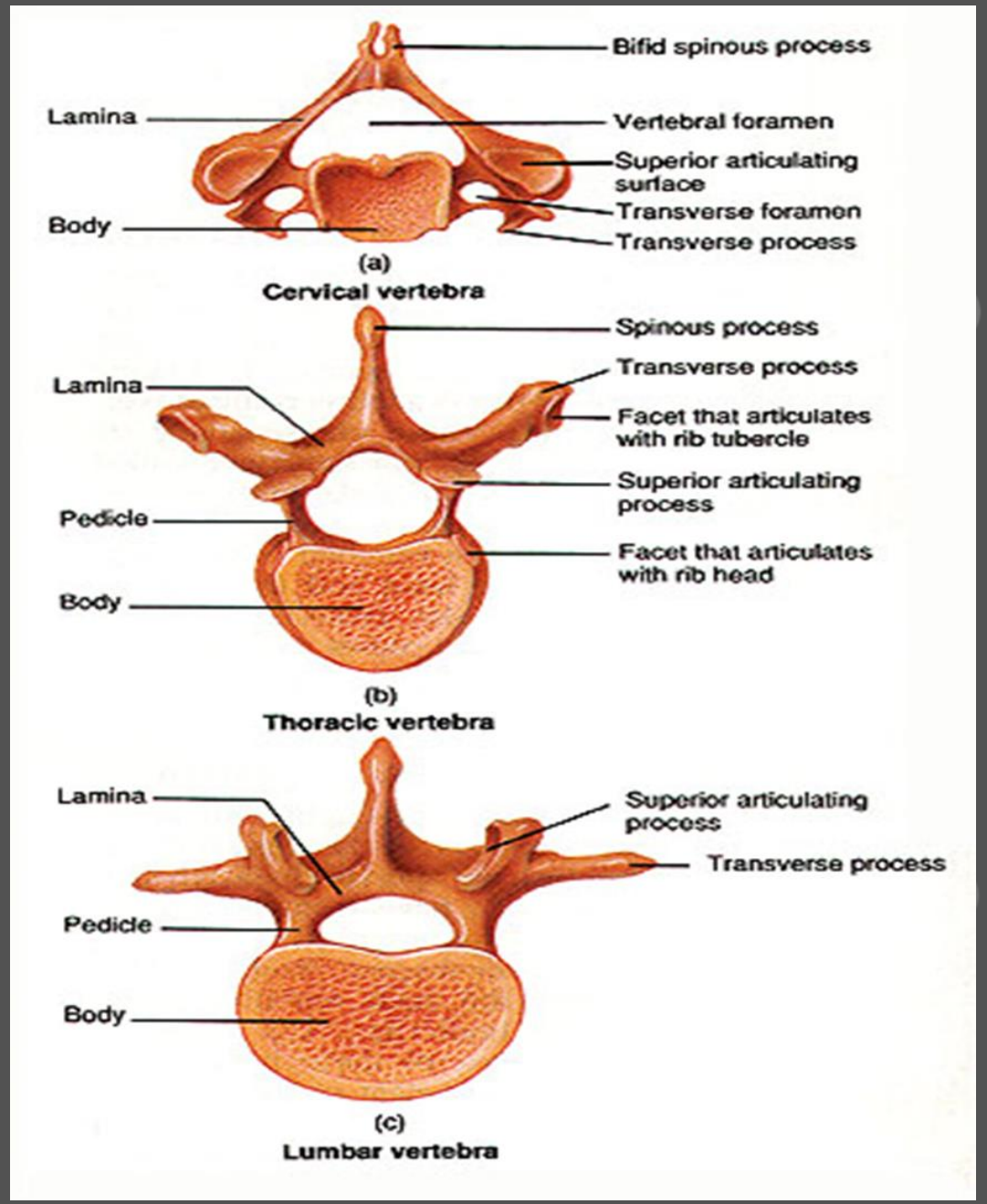
- 7 vertebrae
  - C1-7
- C1 – Atlas
- C2 – Axis
- Atlanto-occipital joint
- Atlantoaxial joint





Cervical vertebrae

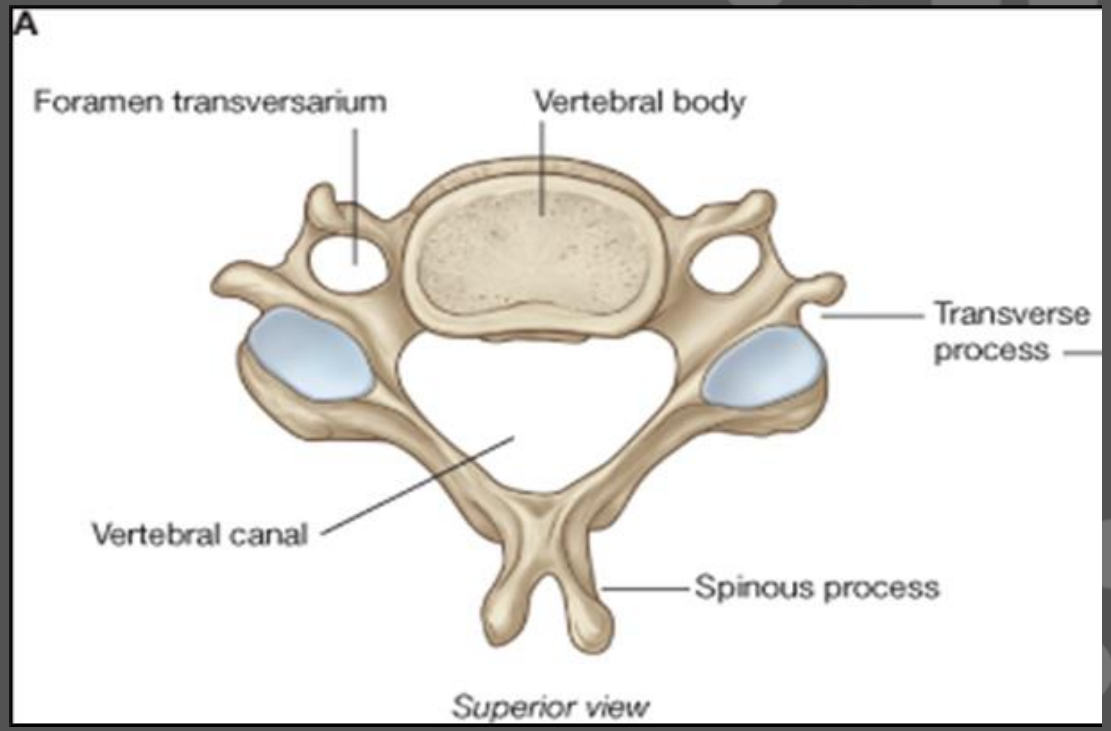
- C3-7 classic vertebrae
- Body is small, oval shaped and short in height
- Large triangular vertebral foramen
- Short pedicle project laterally
- Long narrow laminae



Cervical vertebrae

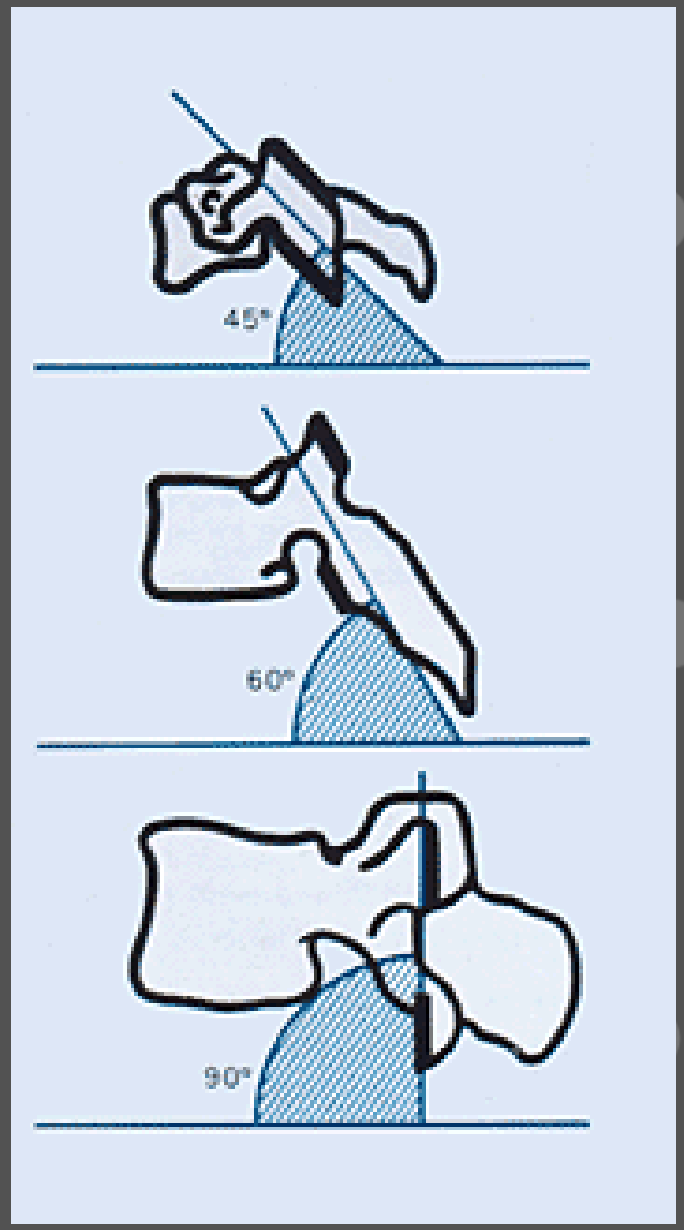
- Short bifid spinous process, muscles attach to nuchal ligament
  - C6-7 long SP
  - C7 rare to be bifid

- Short transverse processes which house foramen transversarium



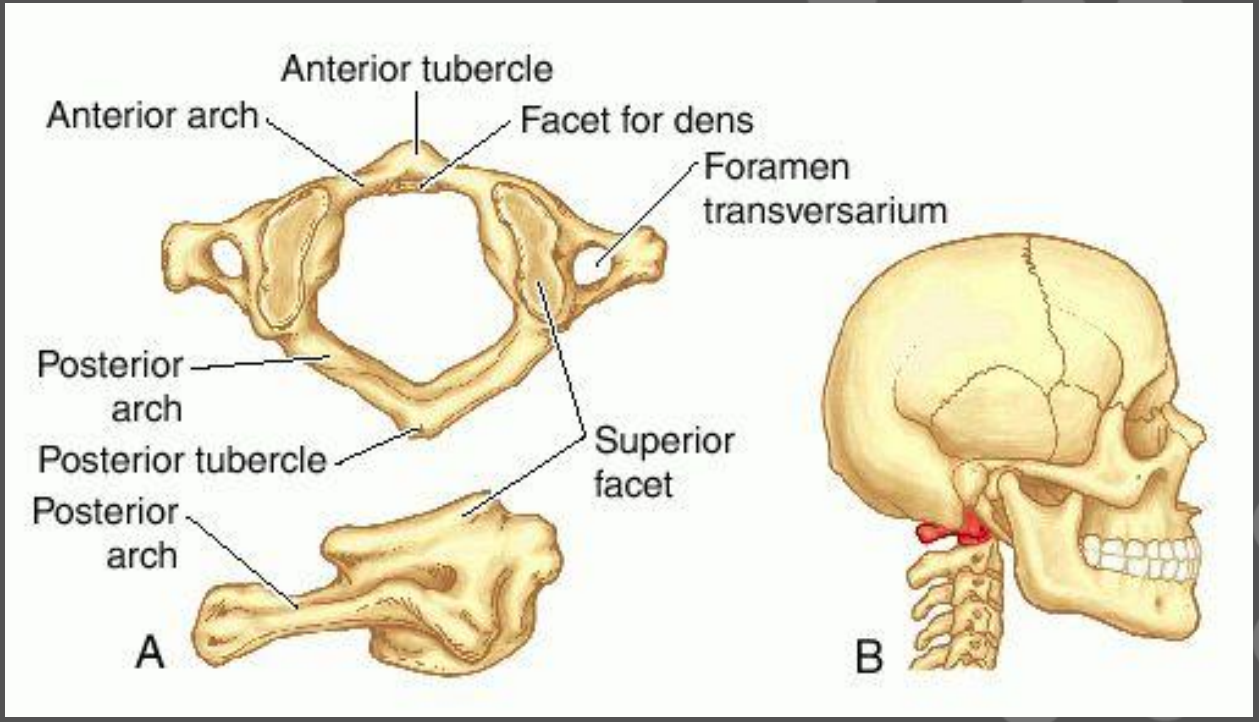
Cervical facet joints

- Facet joints lie at approx. 45° angle
- Superior facet faces superiorly and posteriorly
- Inferior facet faces inferiorly and anteriorly
- Allows for all movements



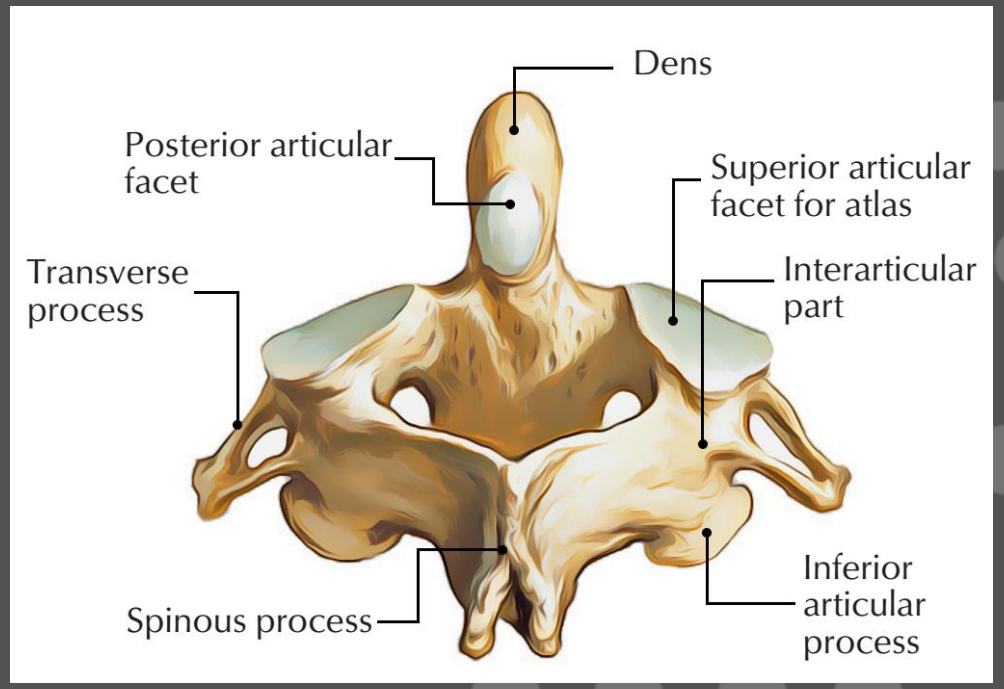
Atlas (C1)

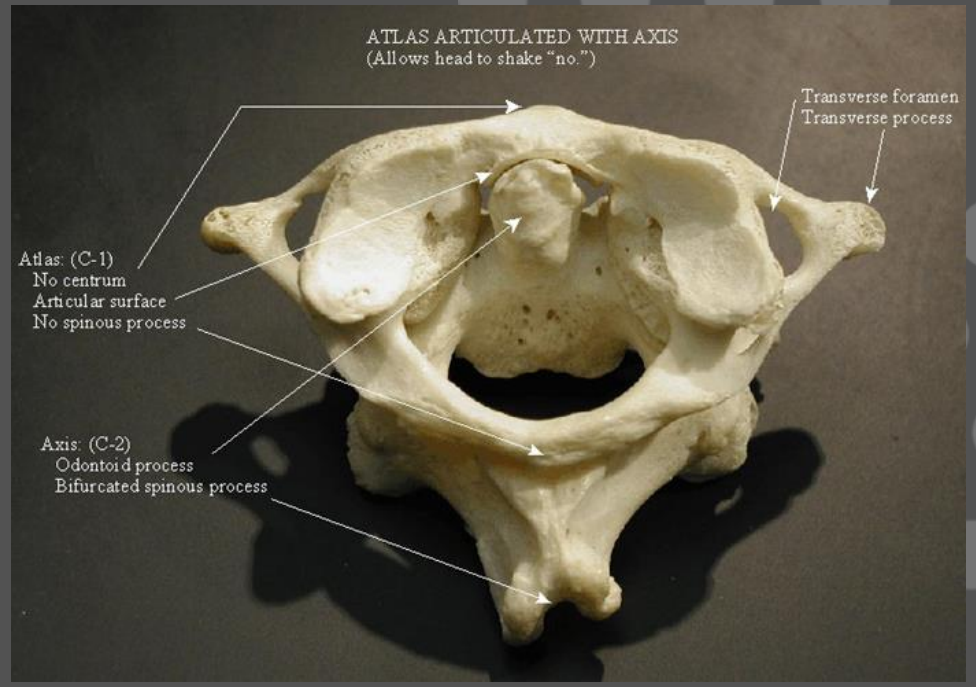
- Articulates with Occiput (atlanto-occipital joint) and with the Axis (atlantoaxial joint)
- No body or spinous process
- Large, wide transverse processes
- Superior facets articulate with occiput
- Anterior arch which articulates with the odontoid process (dens) of C2
- Anterior tubercle origin of anterior longitudinal ligament



Axis (C2)

- Odontoid process (dens) protrudes superiorly and articulates with C1
- Bifid spinous process
- Superior facets support the weight of head allowing dens to articulate freely with Atlas
- Strongest cervical vertebra
- Superior facets face up and slightly laterally





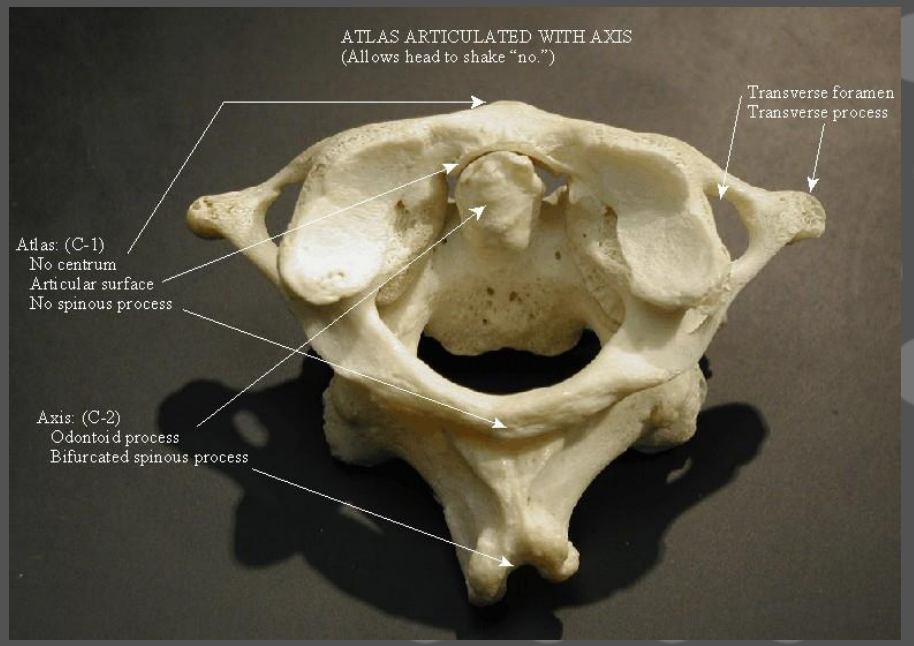
Atlanto-occipital joint

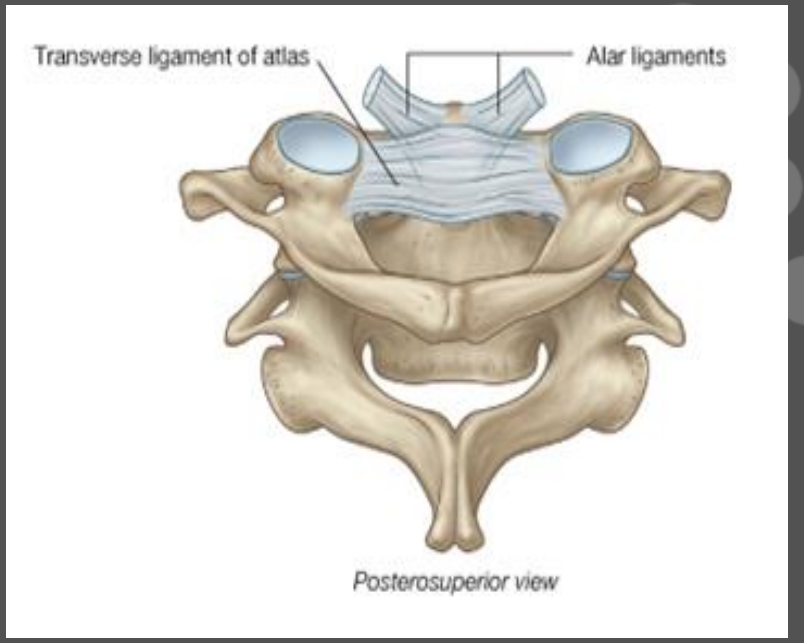
- Articulation of superior facets of C1 and occipital condyle
- Flexion and extension (nodding)
- Lateral flexion
- Very minor rotation from occiput ‘slipping’



Atlanto-axial joint

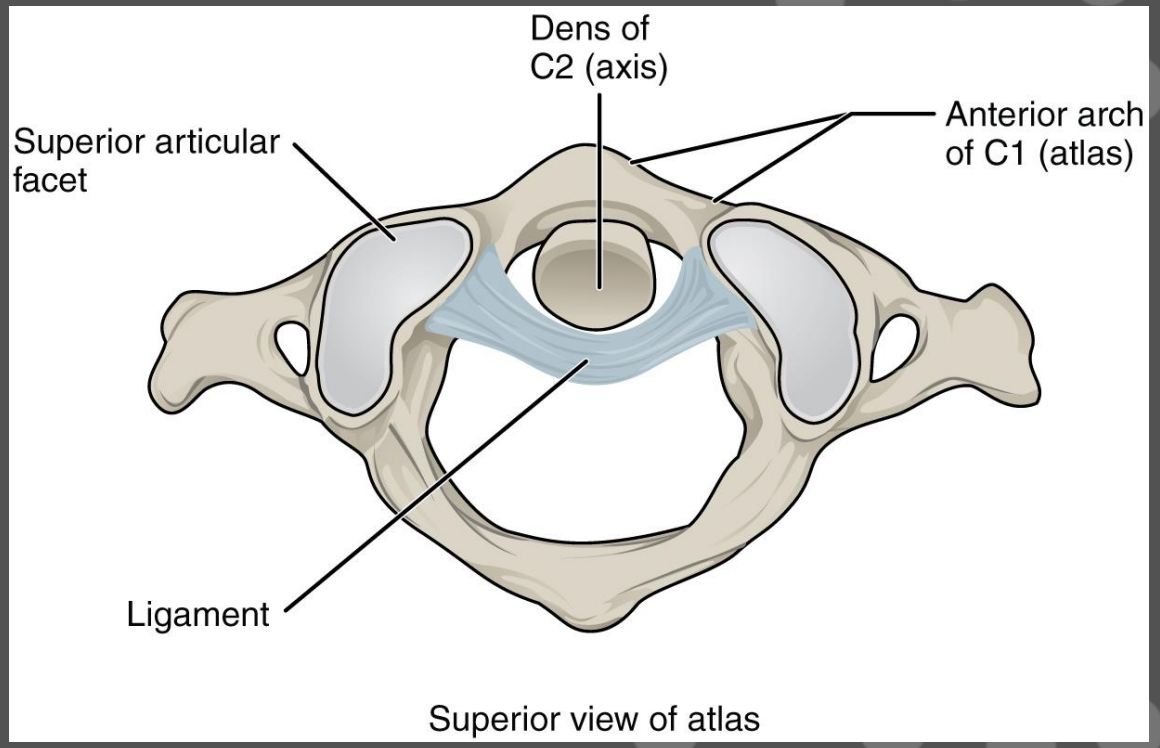
- Lateral and median joints
- Lateral atlanto-axial joint
  - Articulation of inferior facets of C1 and superior facets of C2
- Median atlanto-axial joint
  - Articulation of anterior surface of dens with the posterior surface of the anterior arch of C1
  - Posterior surface of dens with the transverse ligament of C1

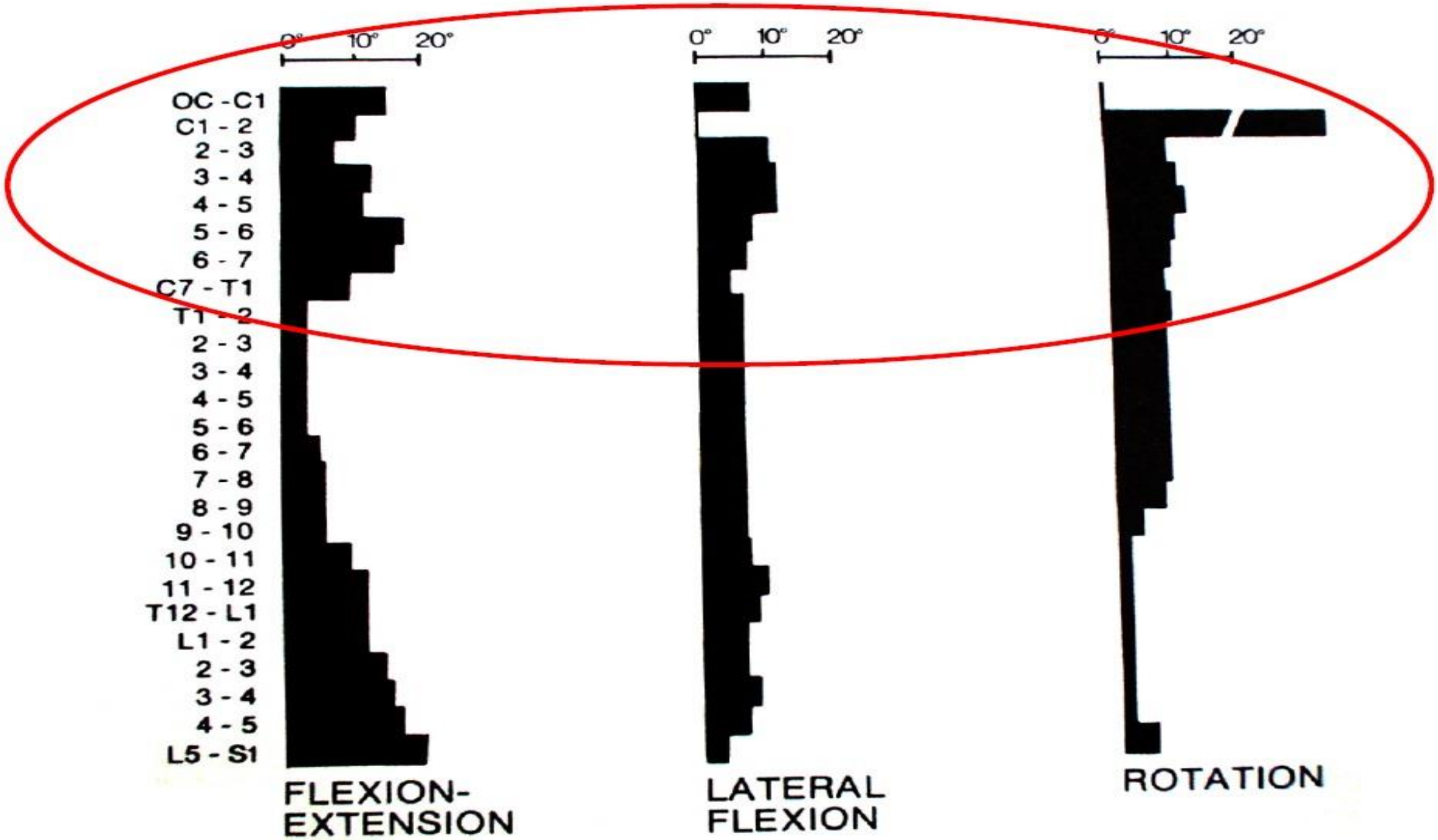




Atlanto-axial joint

- Complex movements due to lateral and medial joints moving together
- Main movement rotation (shaking head to say “no”)
- Atlas spins on it’s axis
- The head and C1 move as a unit on C2




0°
10°
20°
OC - C1
C1 - 2
2 - 3
3 - 4
4 - 5
5 - 6
6 - 7
C7 - T1
T1 - 2
2 - 3
3 - 4
4 - 5
5 - 6
6 - 7
7 - 8
8 - 9
9 - 10
10 - 11
11 - 12
T12 - L1
L1 - 2
2 - 3
3 - 4
4 - 5
L5 - S1
FLEXION-
EXTENSION
LATERAL
FLEXION
ROTATION

Nuchal Ligament

Fibroelastic tissue extending from:

- External Occipital Protuberance to
- Posterior Border of Foramen Magnum to
- Spinous Processes of Cervical Vertebrae

Function:

• **Restricts Hyperflexion:** It acts as a powerful mechanical restraint that limits forward bending of the neck protecting the cervical spine from excessive range of motion.

• **Passive Postural Support:** It supports the weight of the head and provides a vital attachment point for major muscles, allowing for upright stability with minimal active muscular effort.

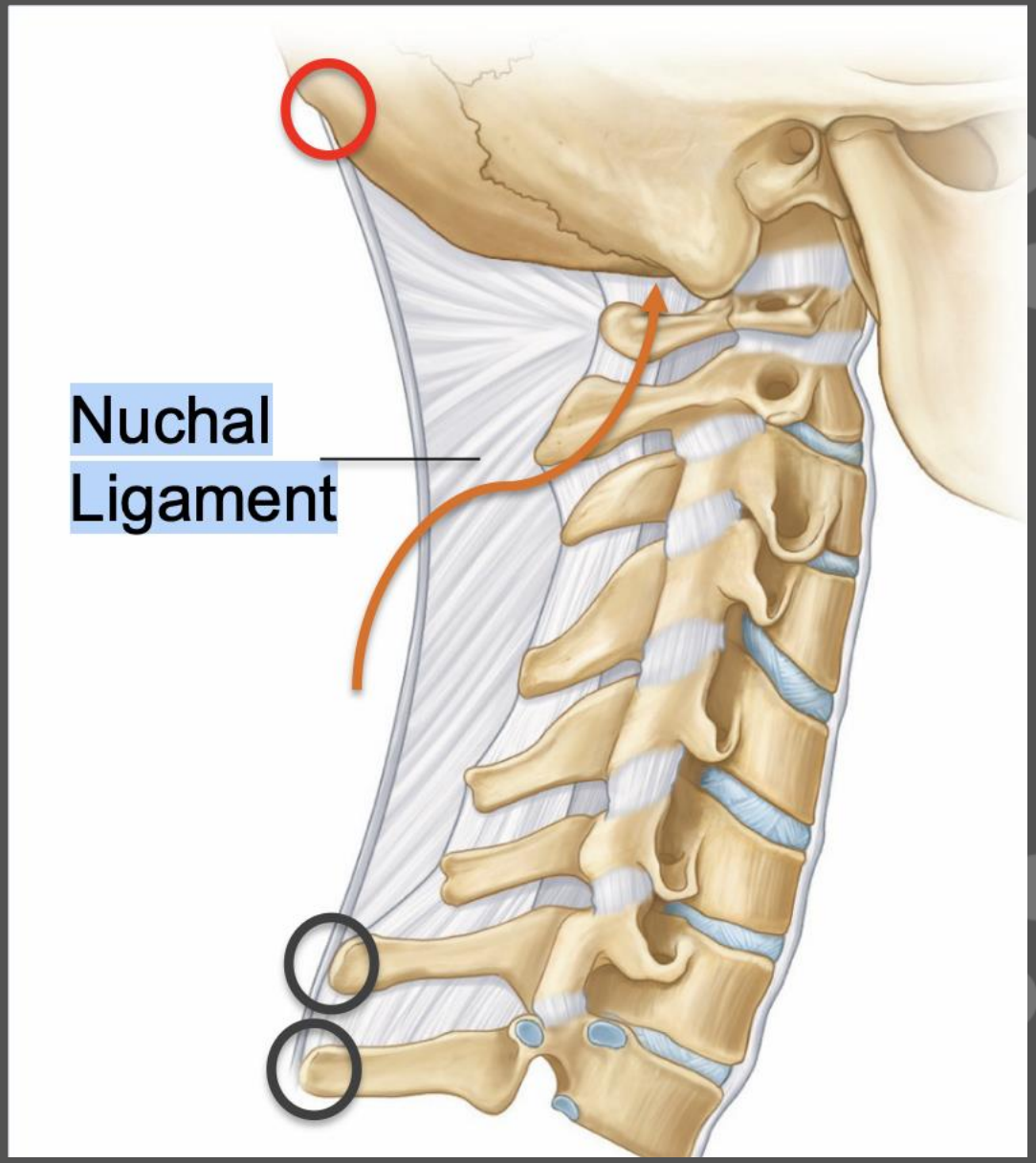


Image from: Moore 8th Ed Fig 2.18

Cervical Muscles

Trapezius

Superficial Posterior Thoracoappendicular Muscle

Innervation: Spinal Accessory Nerve (CN XI)

Action:

- Upper Fibers: Scapula Elevation & Upward Rotation
- Middle Fibers: Scapula Retraction
- Lower Fibers: Scapula Depression & Upward Rotation



Image from:
https://flexikon.doccheck.com/de/Musculus_trapezius

Clinical Pearl
In Upper Crossed Syndrome, the trapezius is split into two opposing roles that drive the postural imbalance: **Upper Trapezius Hyperactivity & Lower/Middle Trapezius Inhibition**

Levator Scapulae

Origin: Transverse processes of C1-C2 vertebrae, posterior tubercles of transverse processes of C3-C4 vertebrae.

Insertion: Superior angle of scapula, medial border of scapula superior to spine of scapula.

Action: Elevates and downwardly rotates pectoral (shoulder) girdle at acromioclavicular and sternoclavicular joints; extends and laterally flexes neck at cervical vertebral joints.

** Clinical Pearl**

In the context of **Upper Crossed Syndrome**, the levator scapulae is a primary culprit in neck and shoulder dysfunction

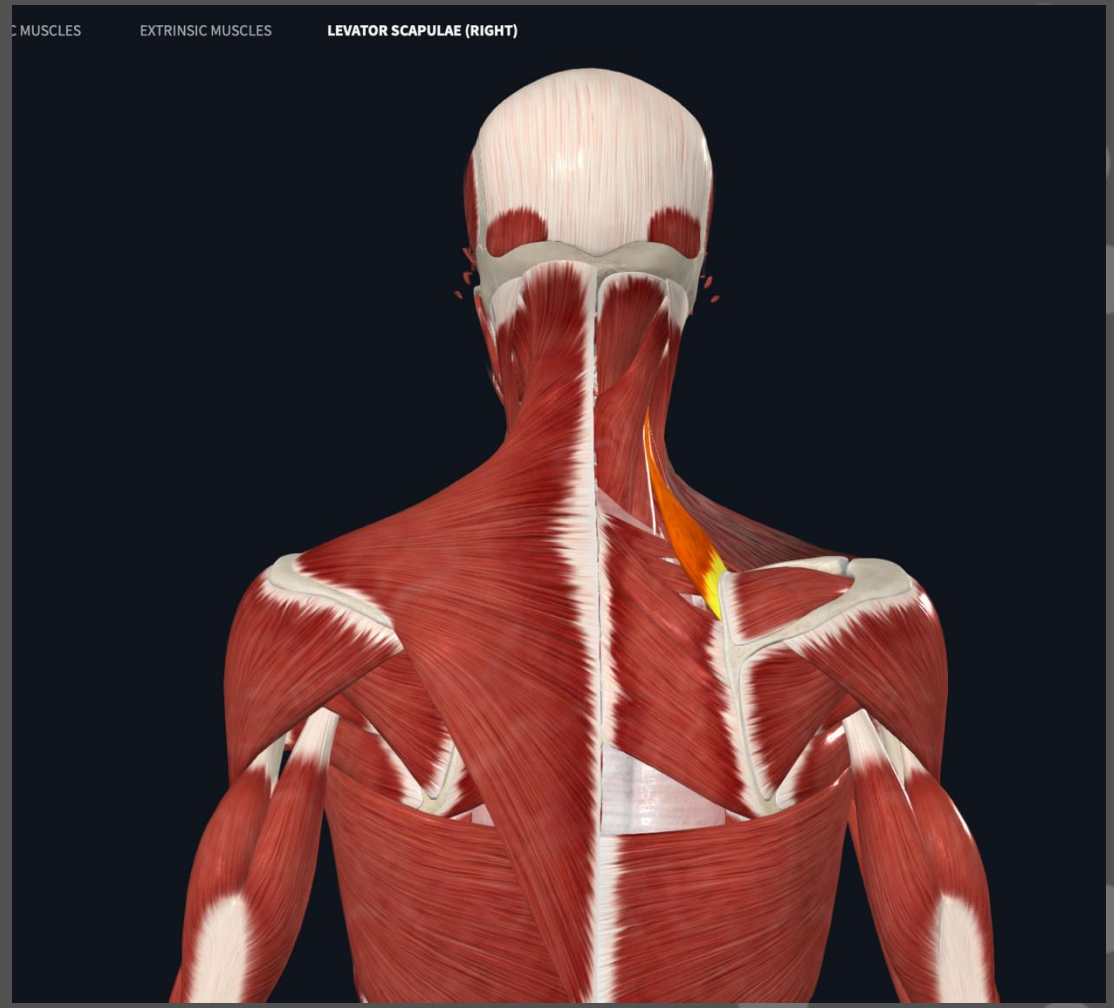


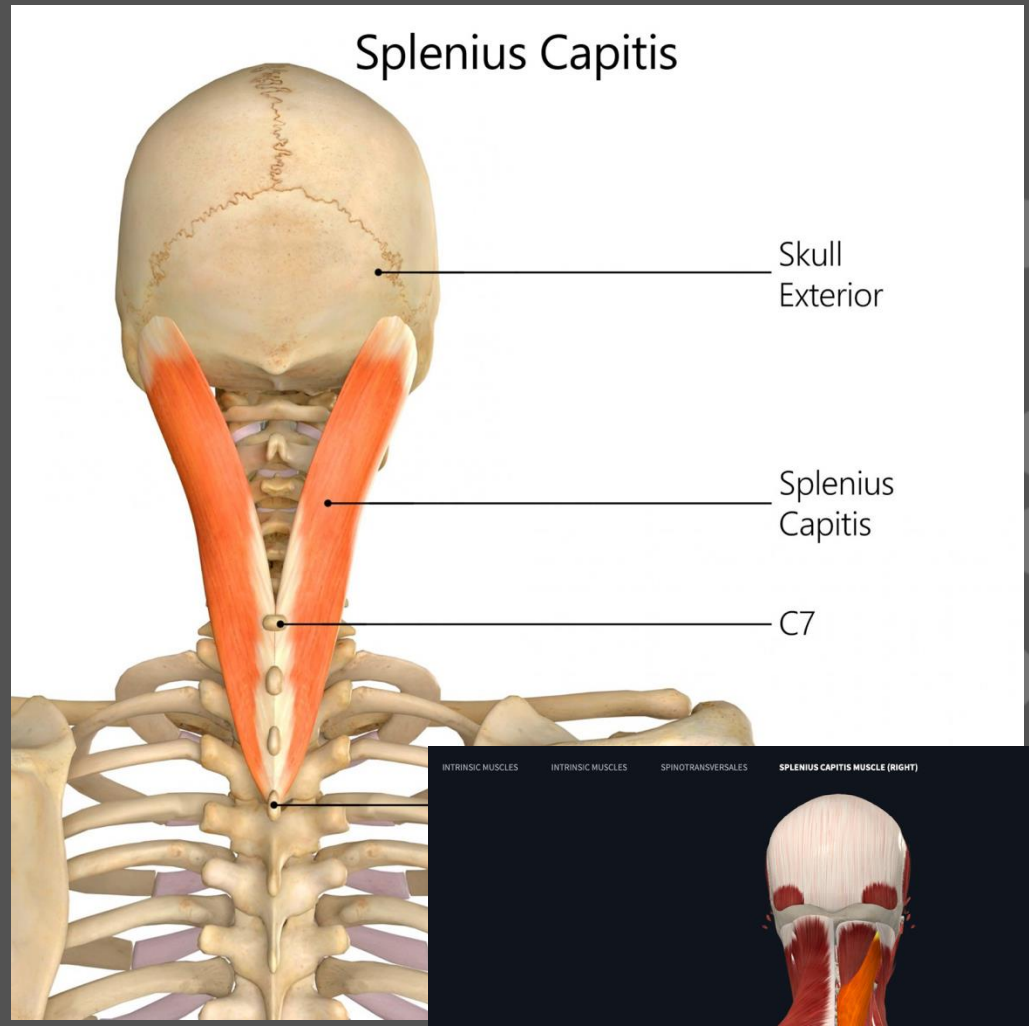
Image from: Complete Anatomy

Splenius Capitis

Origin – Lower ligamentum nuchae, C7-T3 spinous processes

Insertion – Mastoid process, superior nuchal line of occiput

Action – Head extension (bilateral), rotation and lateral flexion (ipsilateral)



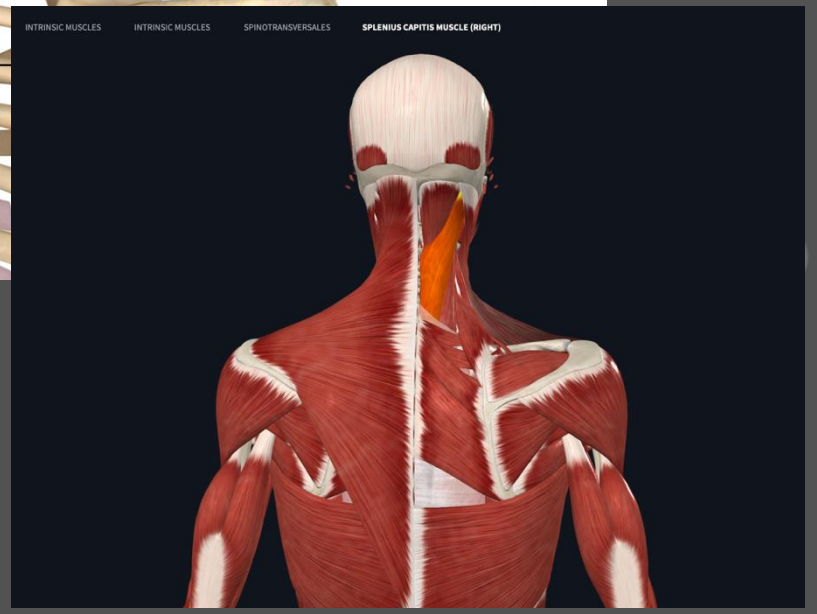


Image from: Complete Anatomy

Semispinalis Capitus

Origin: Superior articular processes of C4-C7 vertebrae; transverse processes of T1-T6 vertebrae.

Insertion: Area located between superior and inferior nuchal lines of occipital bone.

Action: Extends and rotates the head.

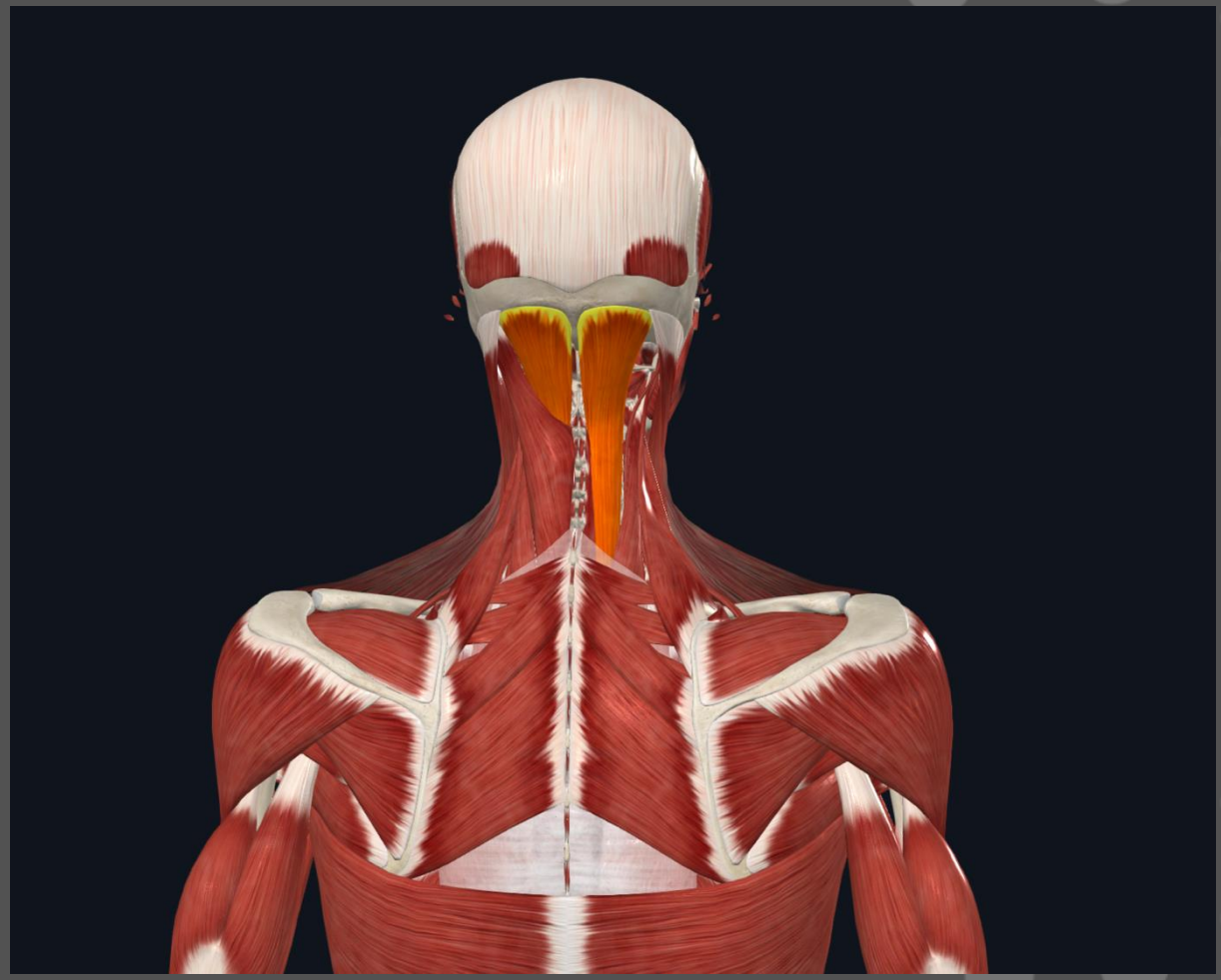
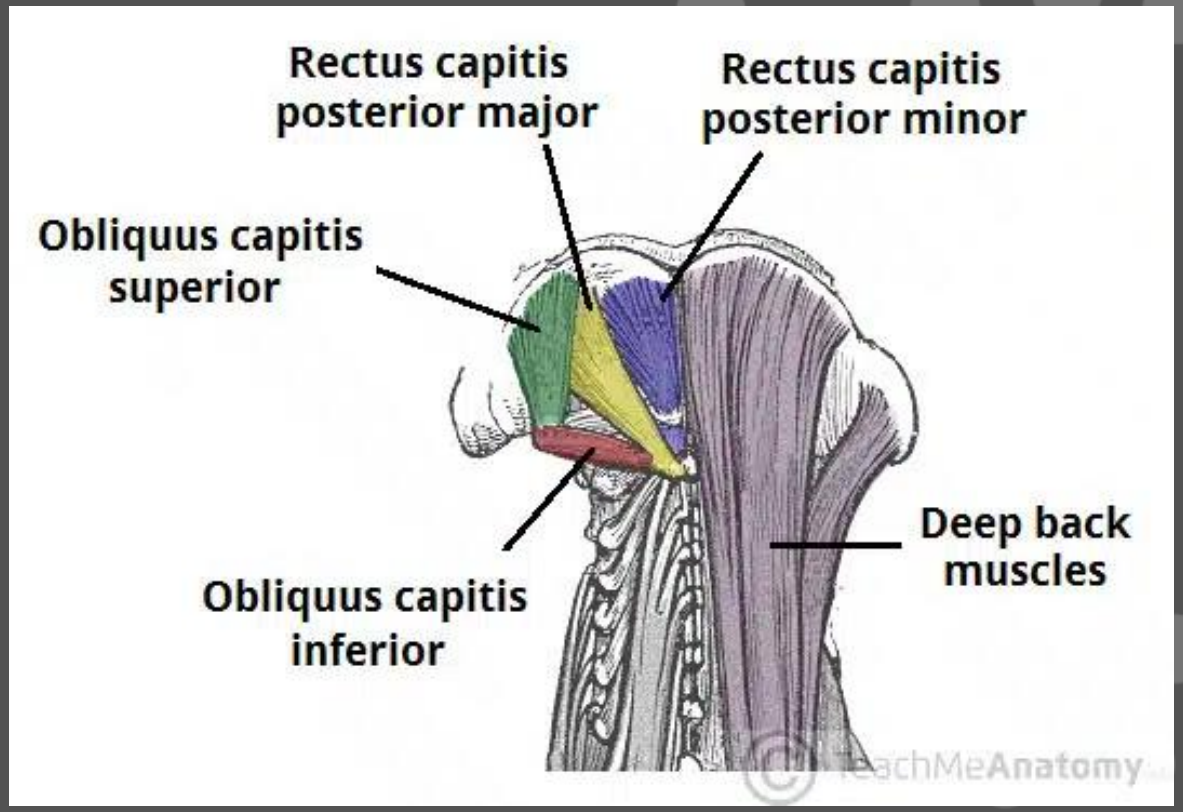


Image from: Complete Anatomy

Suboccipital muscles

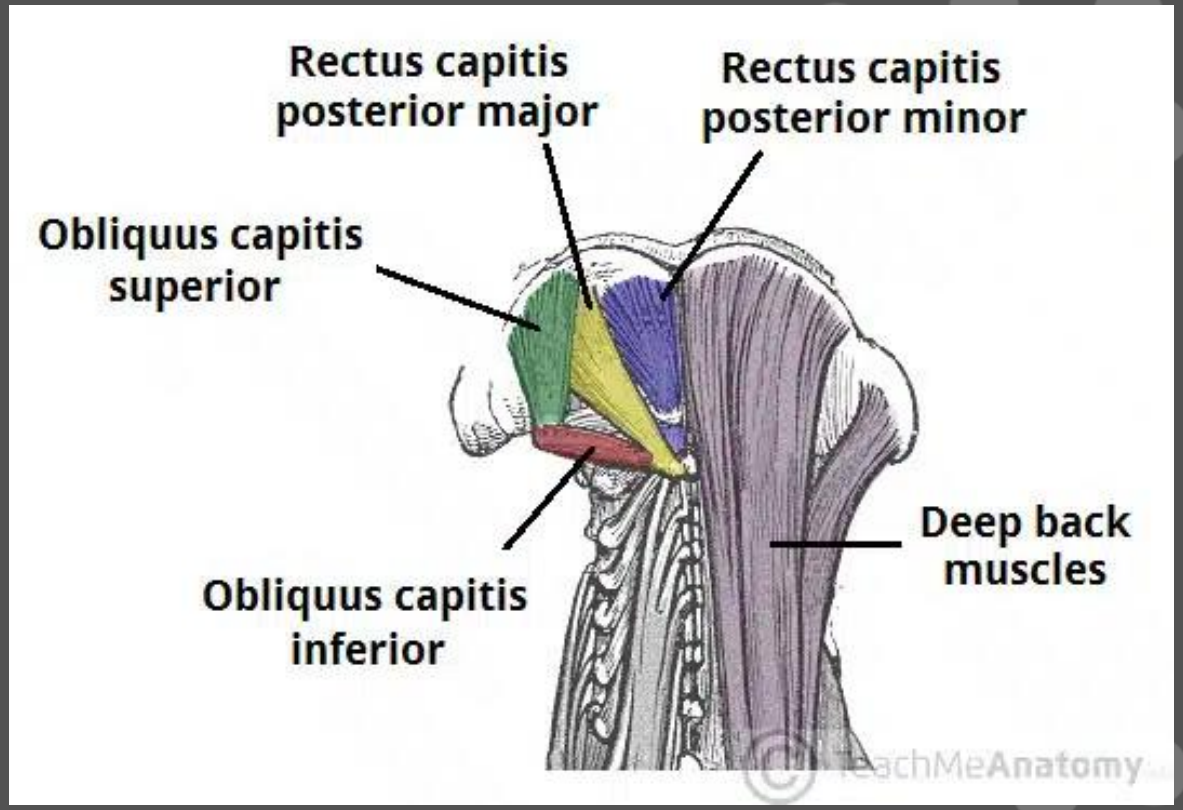
- Rectus capitis posterior major
  - O = C2 spinous process
  - I = Lateral part of nuchal line of occiput
  - A = Extension and rotation of head
- Rectus capitis posterior minor
  - O = C1 posterior tubercle
  - I = Medial part of nuchal line of occiput
  - A = Extension of head



Suboccipital muscles

- Obliquus capitis superior
  - O = C1 transverse process
  - I = Occipital bone between superior and inferior nuchal lines
  - A = Extension of head

- Obliquus capitis inferior
  - O = C2 spinous process
  - I = C1 transverse process
  - A = Extension and rotation of head



****Clinical Pearl****

1. The subocciptals are a Proprioceptive Powerhouse''
2. Rectus capitis posterior minor links to the pain-sensitive lining of the spinal cord, explaining why suboccipital tension frequently triggers cervicogenic headaches.

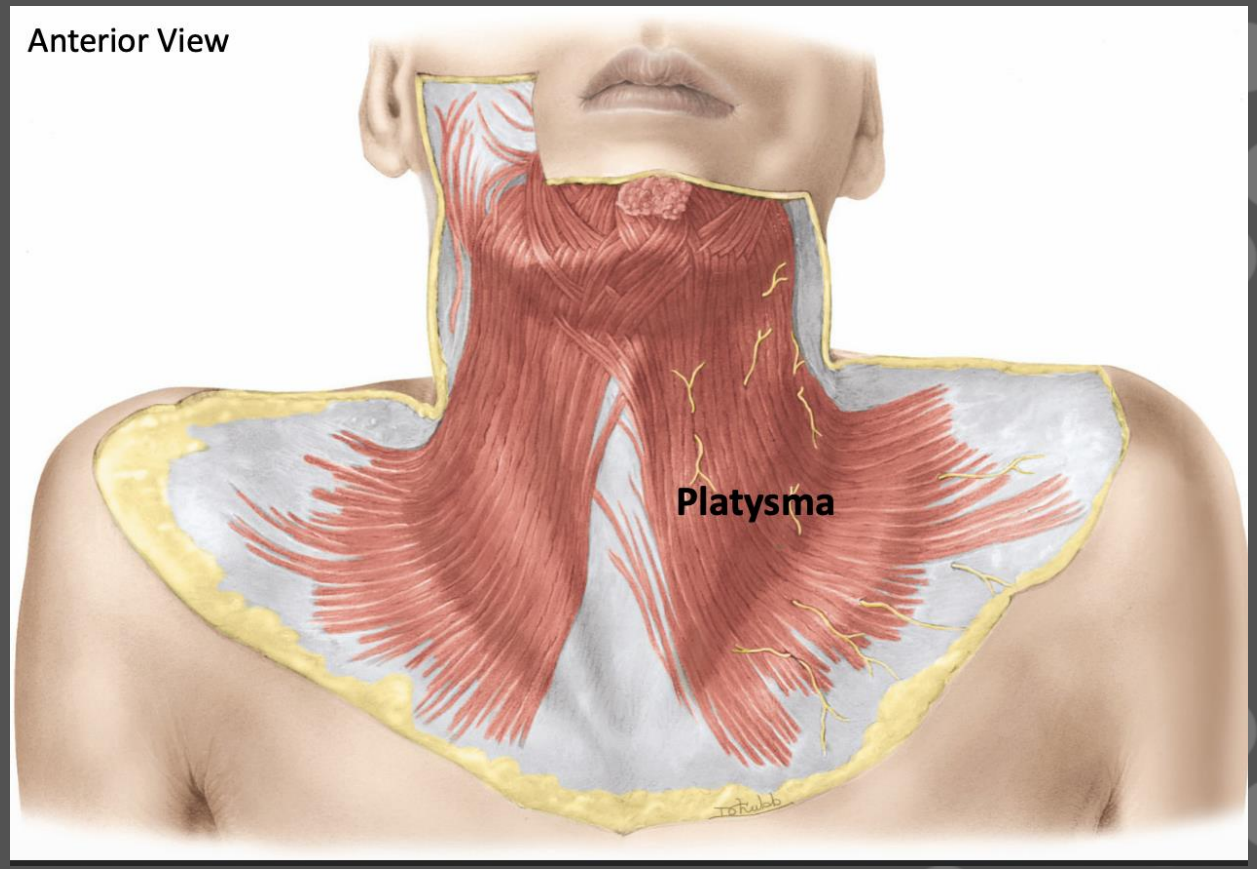
Platysma

Thin layer of subcutaneous tissue that lies between the skin and the Investing Layer of Deep Cervical Fascia.

Innervation: Facial Nerve Anterior View (CN VII)

Action: Draws Corners of the Mouth Inferiorly & Widens It

Draws Skin of the Neck Superiorly When Teeth Are Clenched



Moore 8th Ed Fig 9.5

Sternocleidomastoid

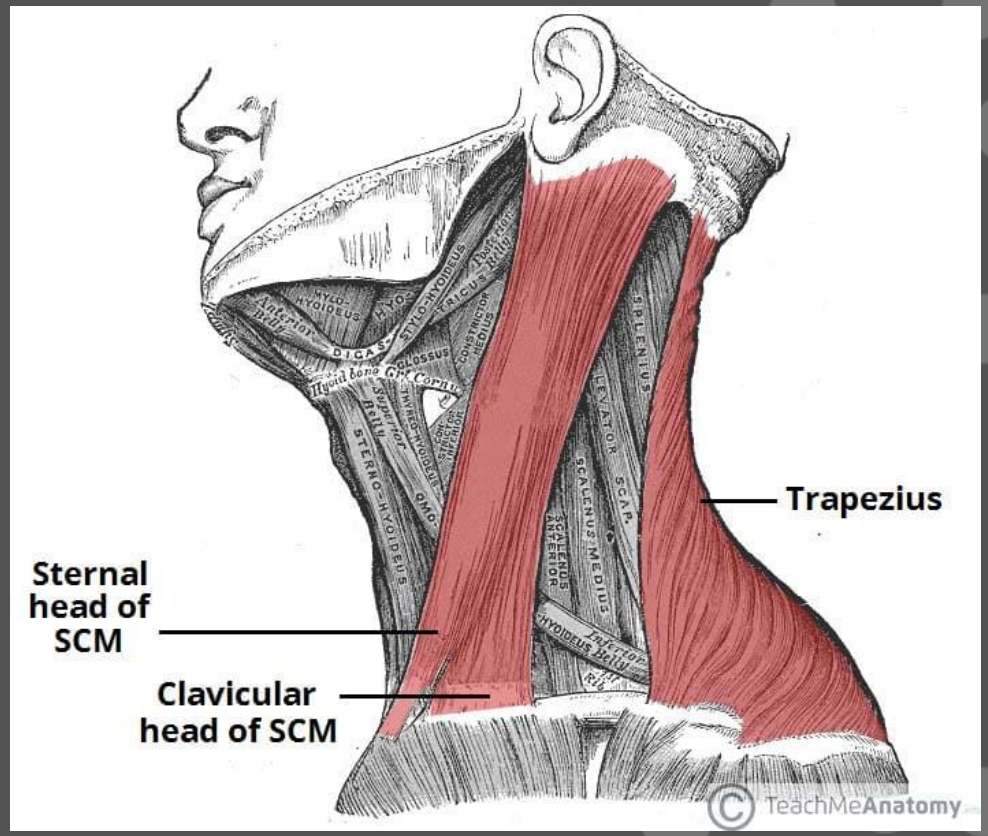
Broad, strap-like muscle that has two heads, Sternal and Clavicular.

Origin – Manubrium of sternum (sternal head), medial 1/3 of clavicle (clavicular head)

Insertion – Mastoid process, superior nuchal line of occiput

Action –

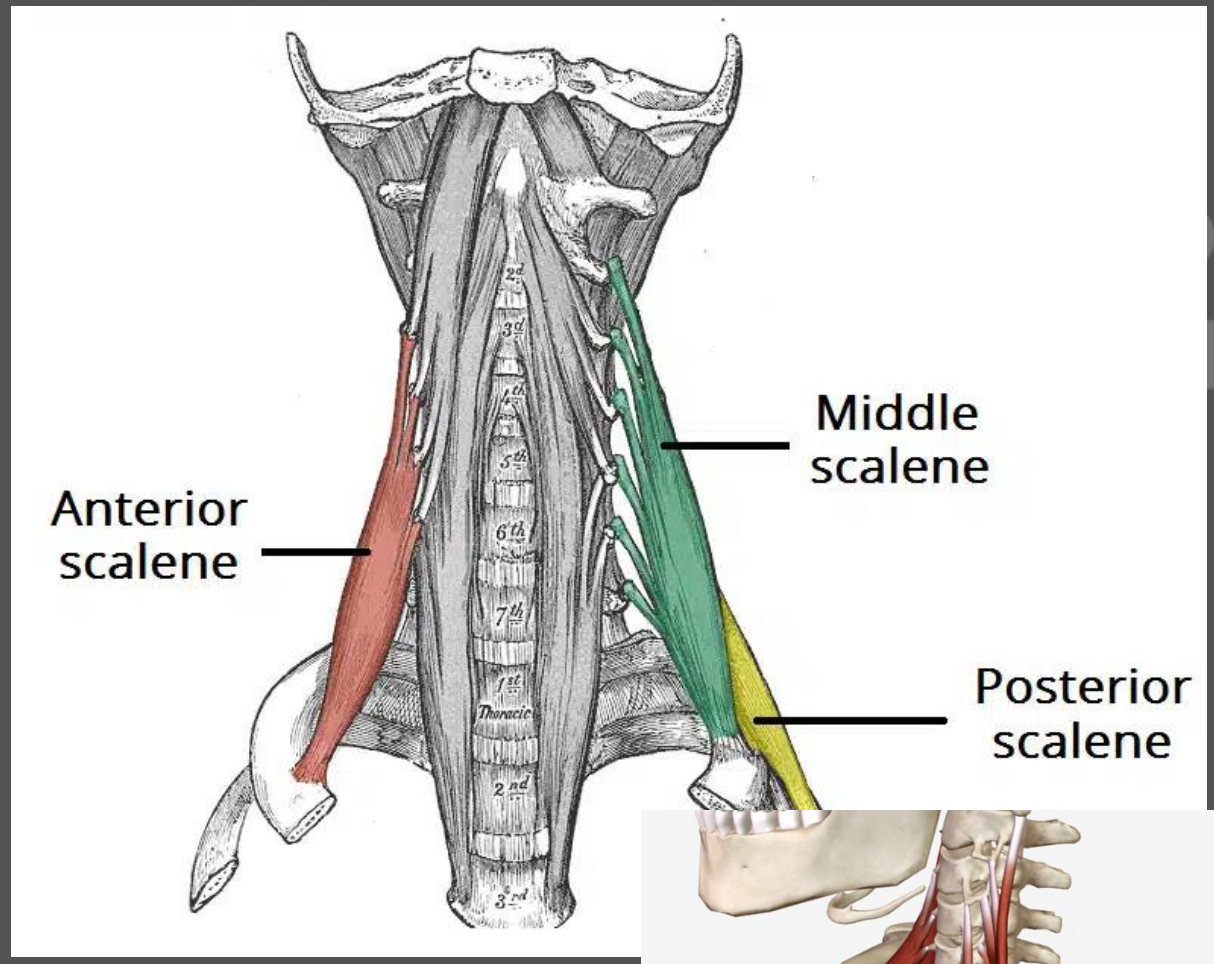
- Unilateral: Head Lateral Flexion with Contralateral Rotation
- Bilateral: Head Extension, Neck Flexion

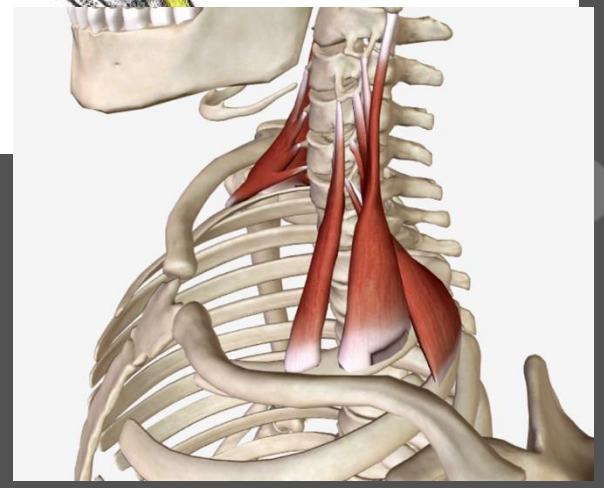


****Clinical Pearl****
Trigger points in the SCM can cause tinnitus, dizziness, and blurred vision without any primary ear or eye pathology, as its tension disrupts proprioceptive input to the vestibular system.

Scalene muscles

- Anterior scalene
  - O = C3-6 transverse processes
  - I = Scalene tubercle of 1st rib
  - A = Elevation of 1st rib, lateral flexion (ipsilateral), flexion (bilateral)
- Middle scalene
  - O = C2-7 transverse processes
  - I = superior aspect of 1st rib
  - A = Elevation of 1st rib, Lateral flexion (ipsilateral)
- Posterior scalene
  - O = C5-7 transverse processes
  - I = 2nd rib
  - A = Elevation of 2nd rib, lateral flexion (ipsilateral)

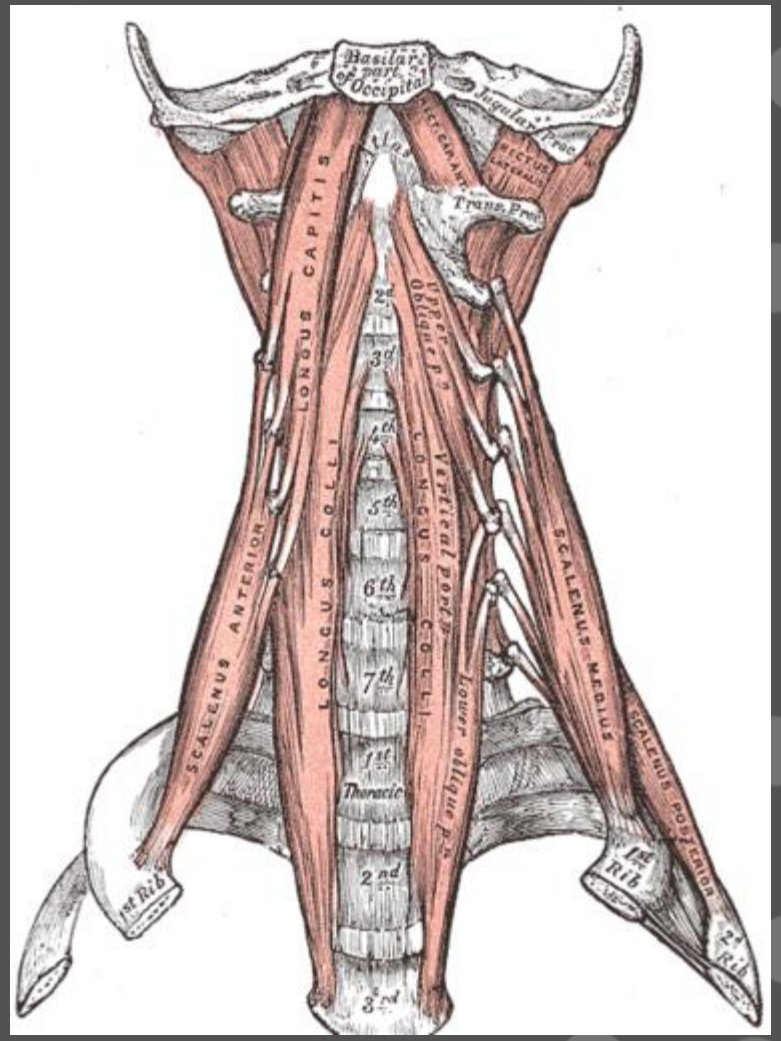




Deep flexors

- Rectus Capitis Anterior
  - O = Anterior surface of lateral mass of C1
  - I = Inferior surface of occiput
  - A = flexion

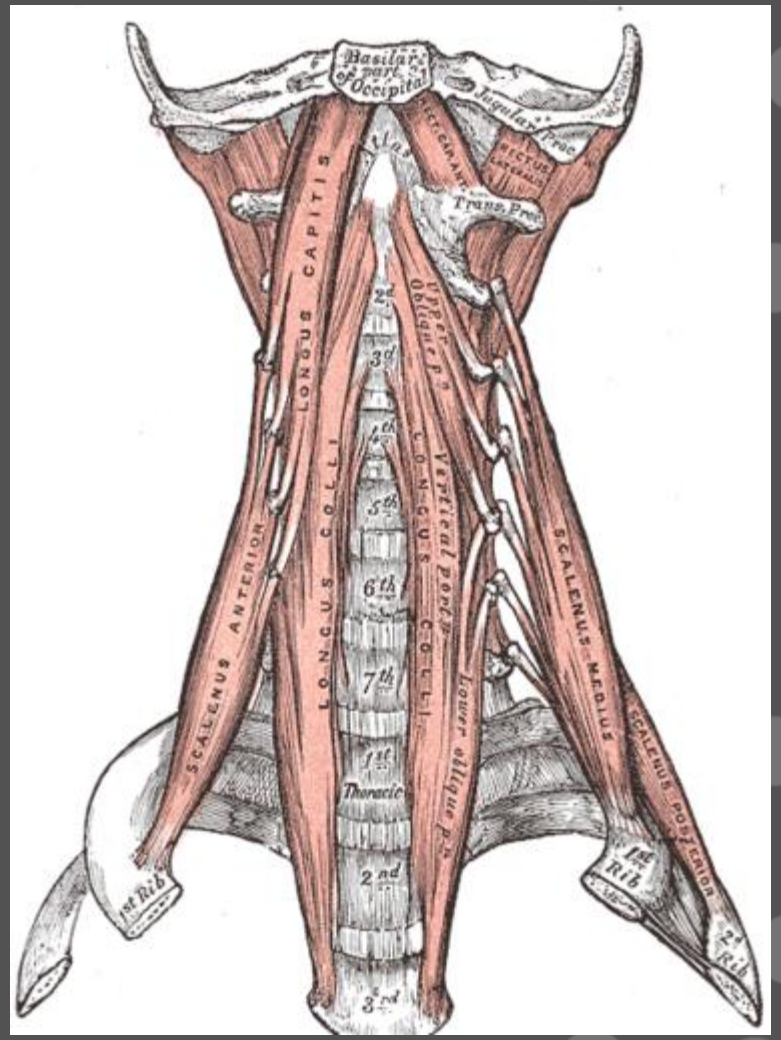
- Rectus Capitis Lateralis
  - O = Transverse process of C1
  - I = Jugular process of occiput
  - A = Stabiisation of head and lateral flexion



Deep flexors

- Longus Capitis
  - O = Anterior tubercles of TP of C3-6
  - I = Basilar portion of occiput
  - A = Flexion (bilateral), ipsilateral rotation/lateral flexion (unilateral)

- Longus Colli / Longus Cervicis
  - O = Anterior surfaces of C3-T3
  - I = Anterior arch of C1, anterior tubercles of C5-6, anterior surfaces of C2-4
  - A = Flexion (bilateral), ipsilateral lateral flexion and rotation (unilateral)



Hyoids

Suprahyoid

- Superior to the Hyoid bone, connecting it to the Skull. Form the floor of the Mouth.
- primarily elevate the hyoid and floor of the mouth for swallowing and speech
- Made of 4 muscles in total

Infrahyiod

- Inferior to the Hyoid bone.
- Depress the hyoid to stabilize the larynx, collectively crucial for swallowing and respiration.
- Also made of 4 muscles total

****Clinical Pearl****

Tension across these muscles can lead to vocal fatigue and a "thin" or "pinched" tone.

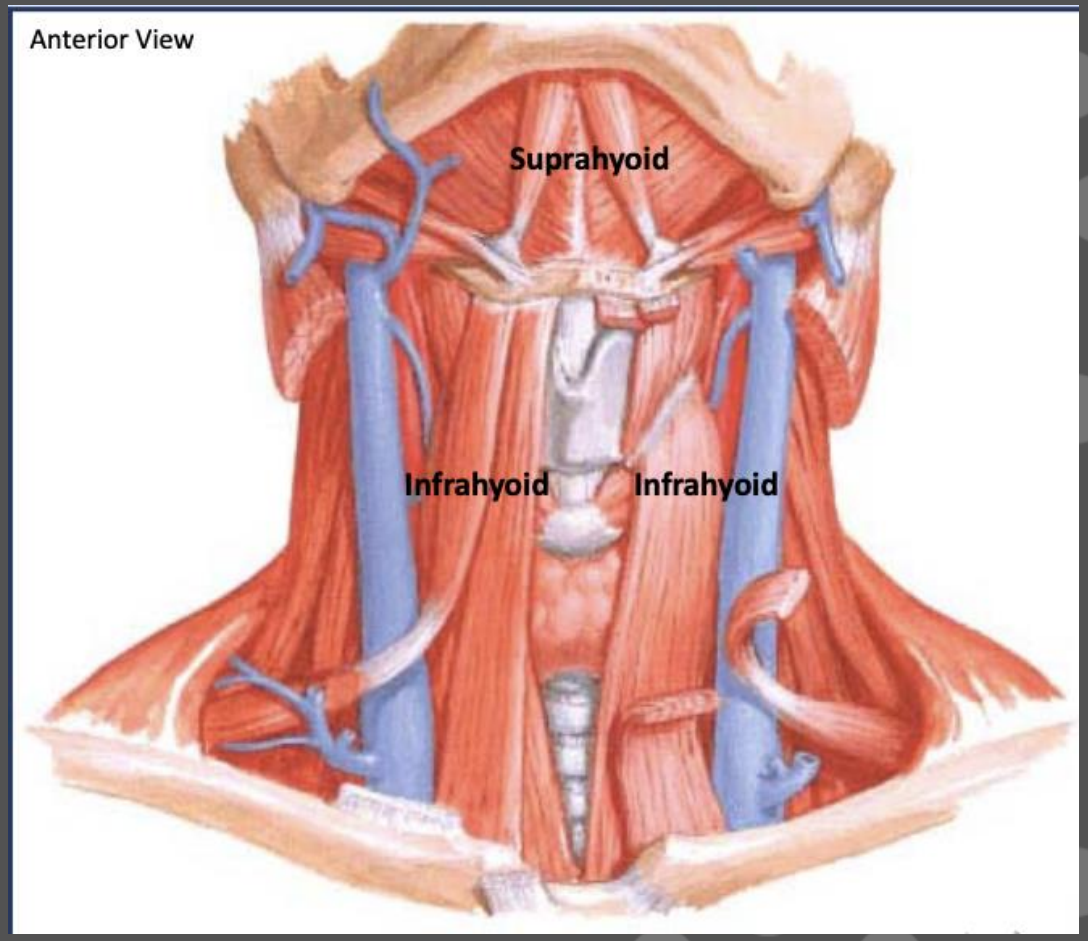


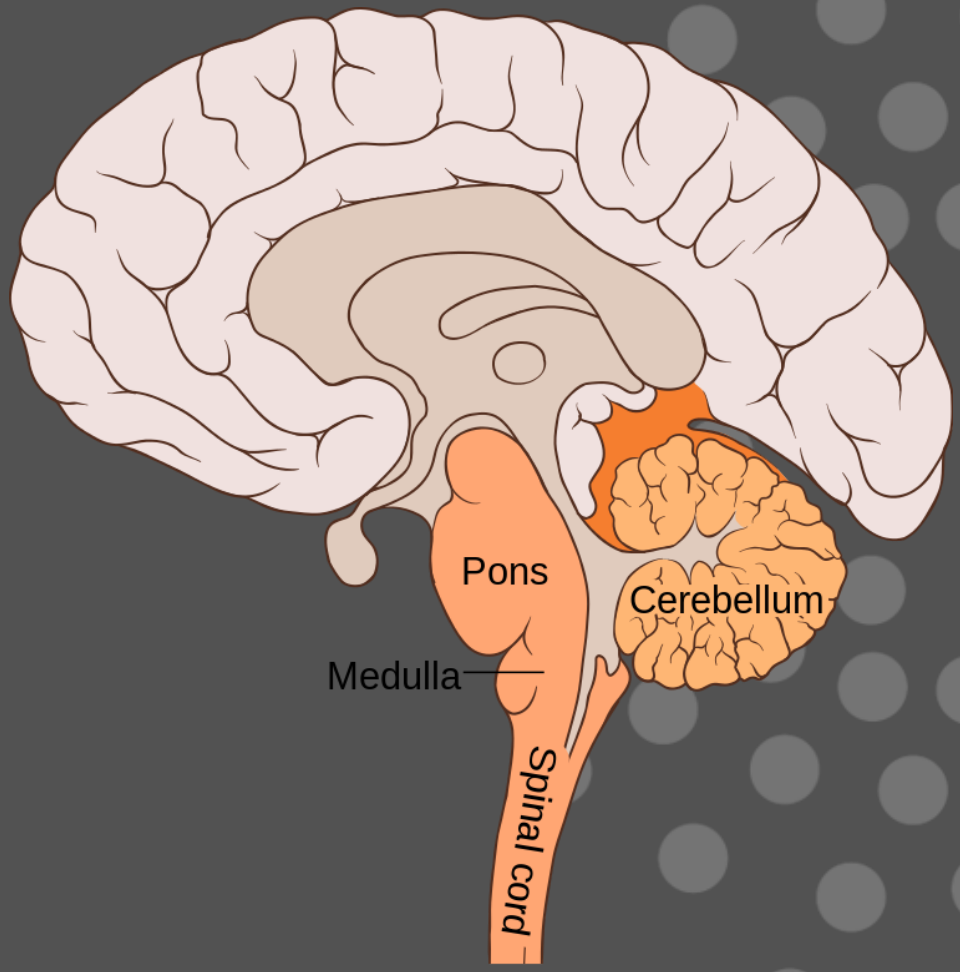
Image from: Netter 7th Plate 35

Quick Major Clinical Consideration

Vertebrobasilar System

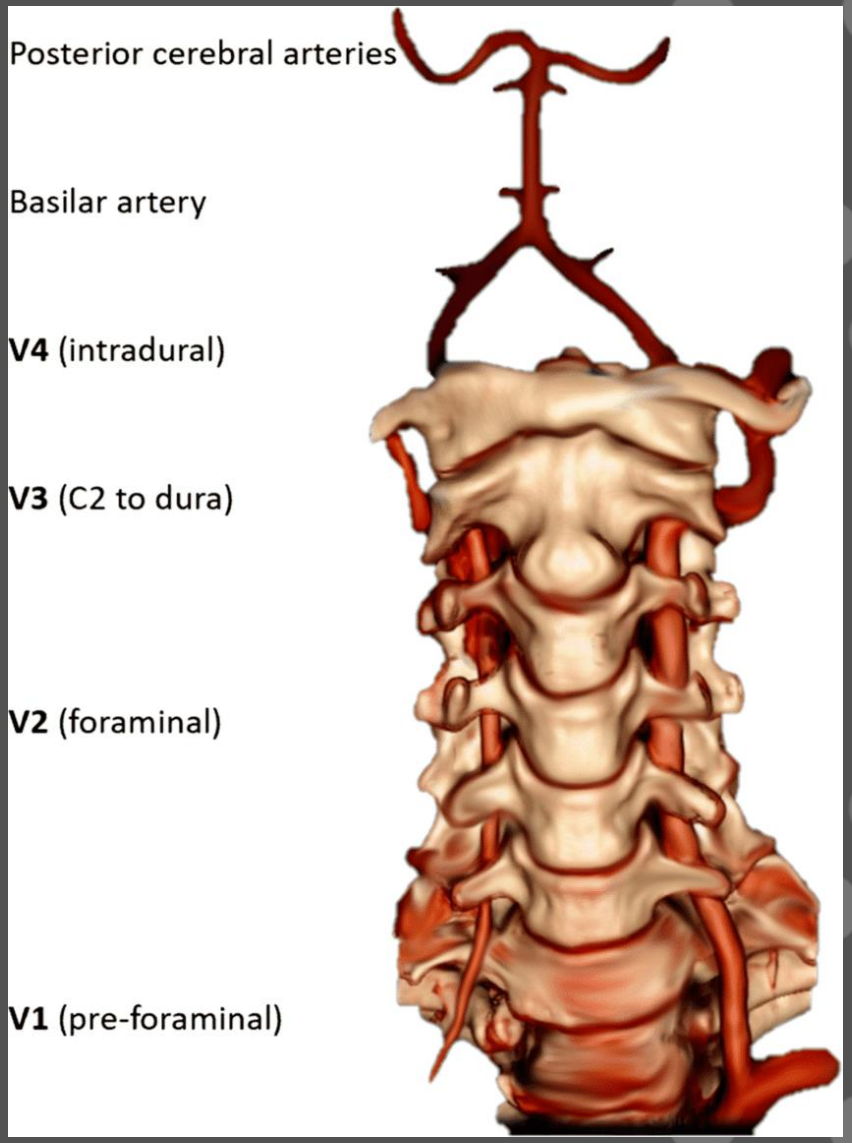
- Vertebrobasilar artery mainly supplies the hindbrain which is responsible for balance, communication, co-ordination

- Can be restricted due to too much movement (rotation) – vertebrobasilar insufficiency



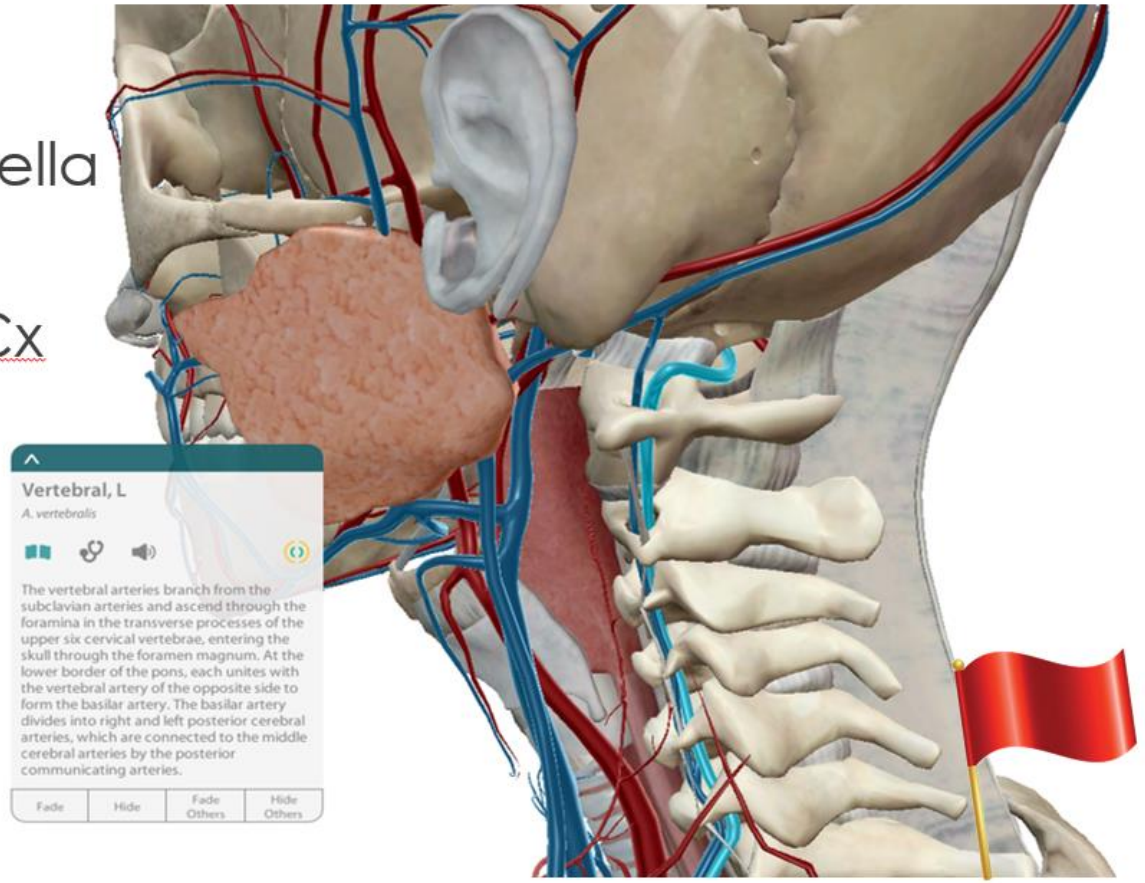
Vertebrobasilar System

- Composed of Subclavian Artery, Vertebral Artery and Basilar Artery
- Vertebral artery runs through foreman transversarium of Cx and then meets the contralateral VA to form the Basilar Artery
- VA kinks between C2 and C1



Vertebrobasilar Artery Insufficiency

- Supplies 20% of blood to brain – predominantly brain stem, cerebella and occipital lobes (Kerry, 2000)
- Blood flow can be restricted by Cx movement
- Travels through foramen transversarium of Cx vertebrae
- Can cause strokes / TIA due to ischemia

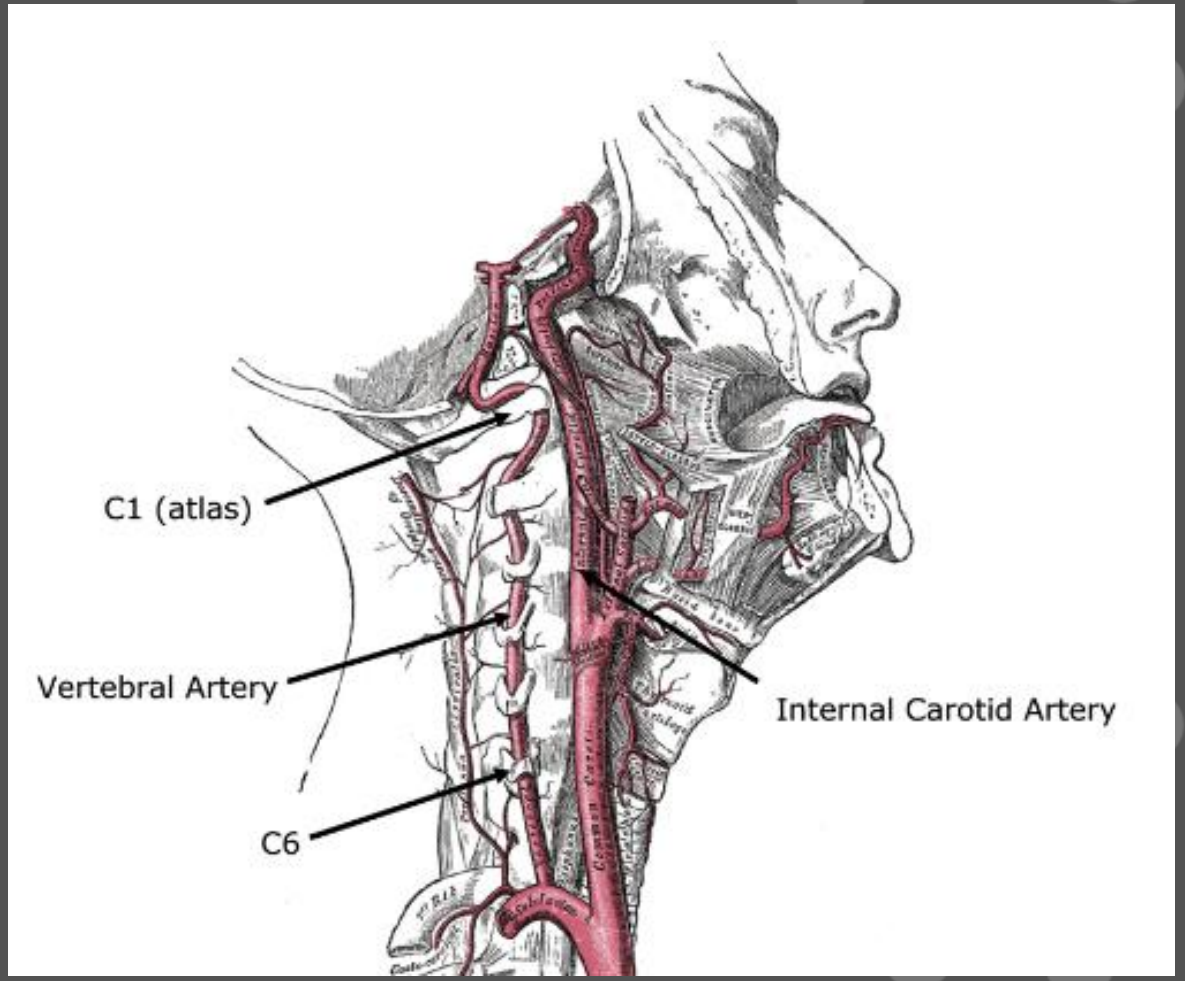


Signs and symptoms of VBI

- 5 D's & 3 N's

Drop attacks
Dizziness
Dysphasia – difficulty swallowing
Dysathria – difficulty speaking
Diplopia – Blurred vision

Nystagmus – uncontrolled eye movement
Nausea
Neurological symptoms



So What? How to screen your patient

If any of the following signs are positive, pre-manipulation or if you are going to be doing extreme extension or extreme rotation with your patient, be sure to clear the arty by doing the **The Vertebral Artery Test (VAT) / Extension-Rotation Test.**

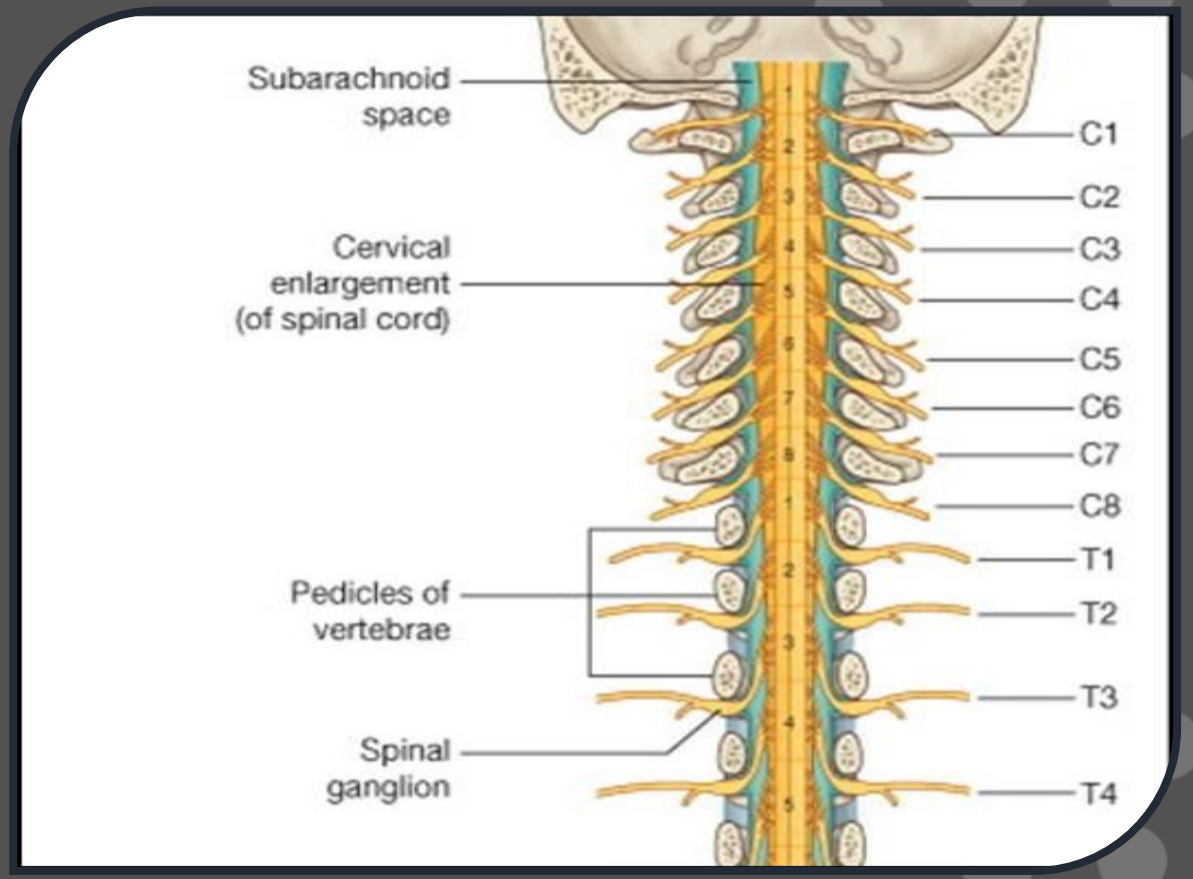
PhysioTutors VBI test: https://www.youtube.com/watch?v=V49zvZMZQTM



The Nervous System

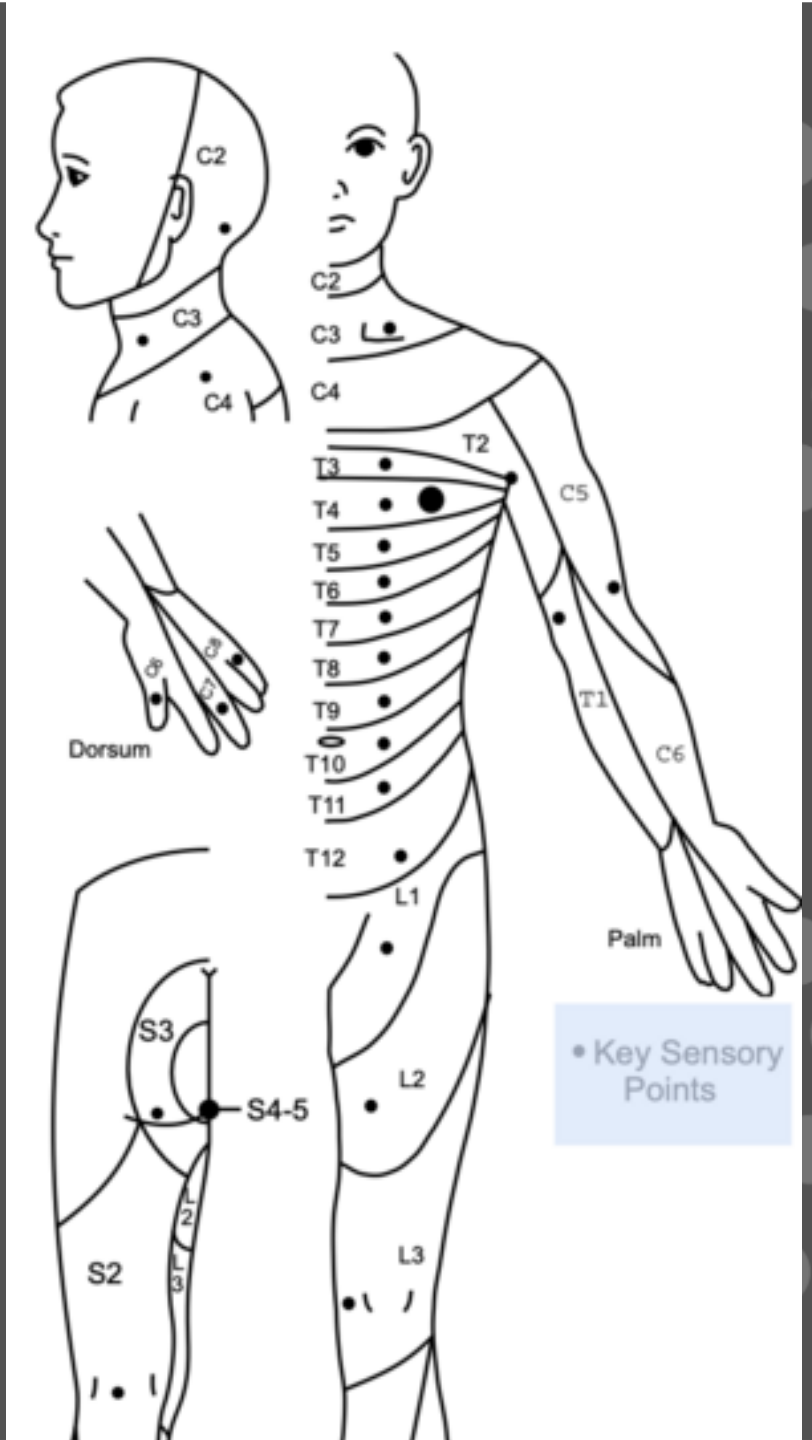
Nerve roots

- Lx = 5 vertebrae and 5 nerve roots
- Tx = 12 vertebrae and 12 nerve roots
- Cx = 7 vertebrae and 8 nerve roots



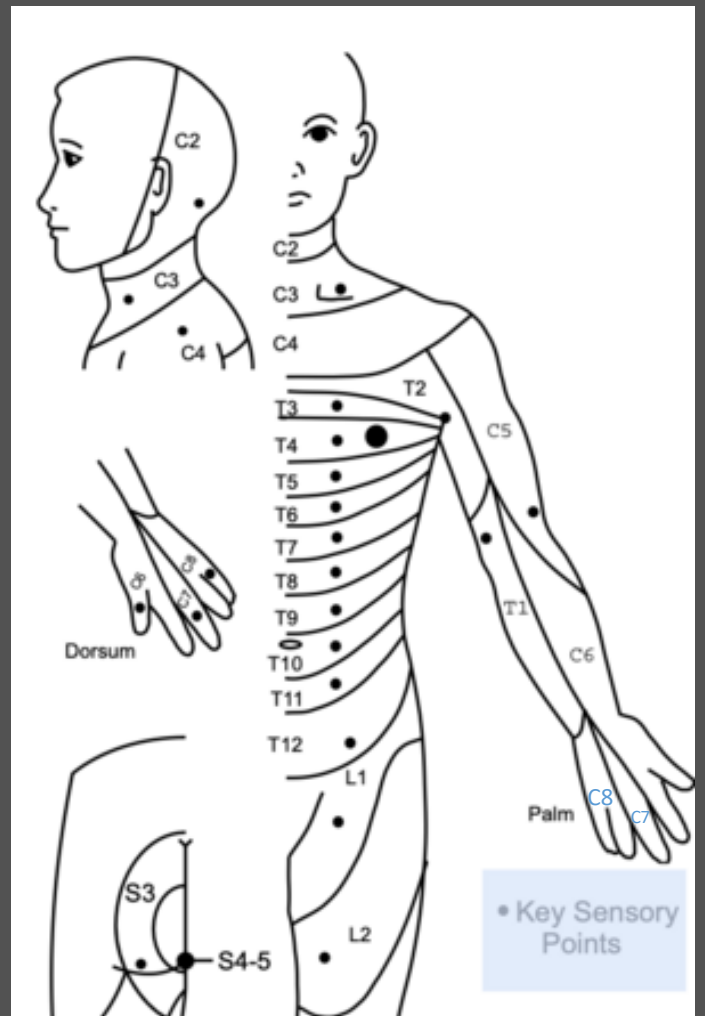
Nerve roots

- A **dermatome** is a specific area of skin supplied by the sensory fibers of a single spinal nerve root, whereas a **myotome** is a group of muscles innervated by the motor fibers of that same spinal nerve root.
- Testing these allows a clinician to localize the level of a nerve root lesion; for example, diminished sensation at the lateral thumb (**C6 dermatome**) paired with weak wrist extension (**C6 myotome**) strongly suggests a compromise at the C6 spinal level.



Nerve roots

- Upper Limb Dermatomes:

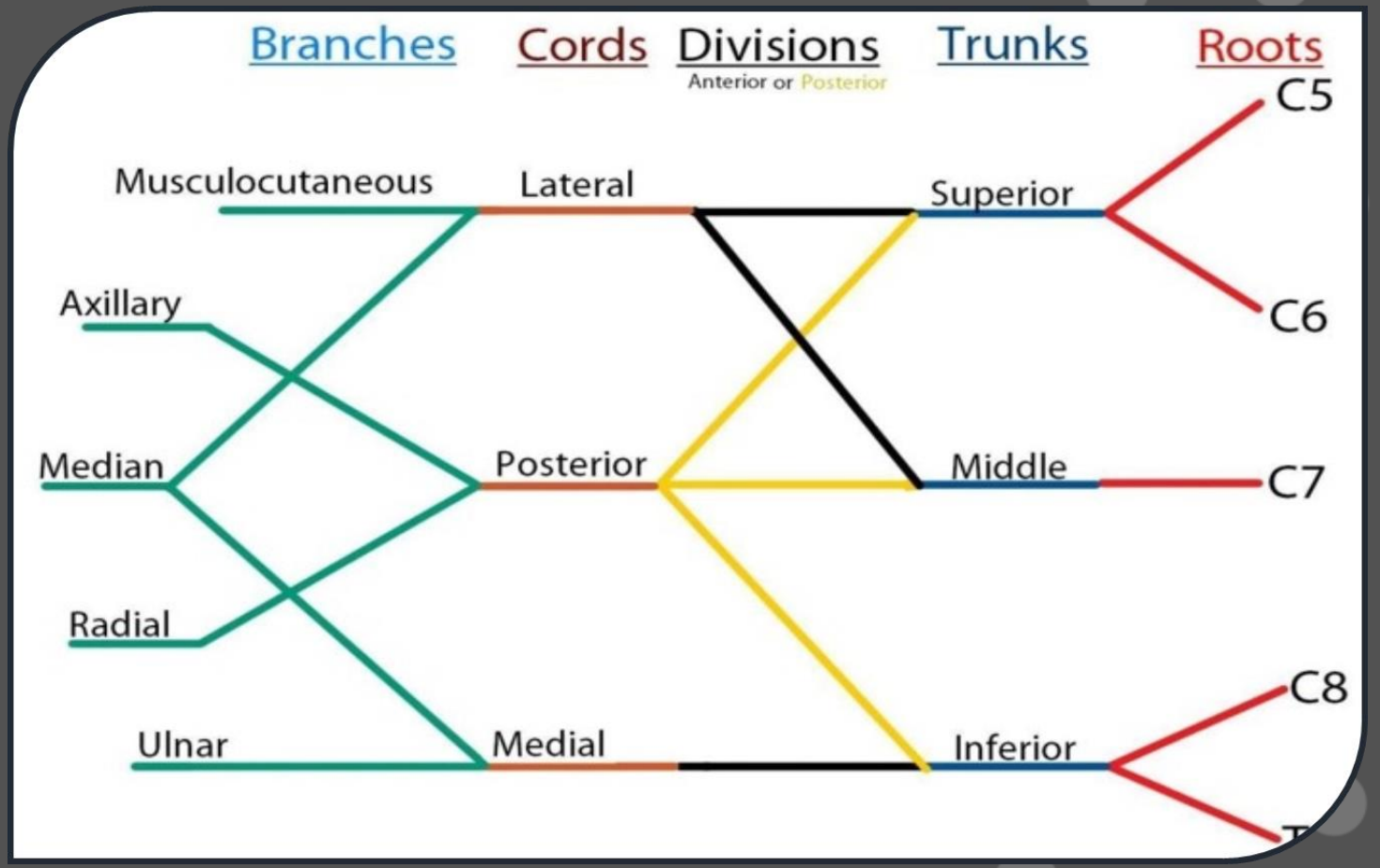


- Upper Limb Myotomes:
  - C5
  - C6
  - C7
  - C8
  - T1

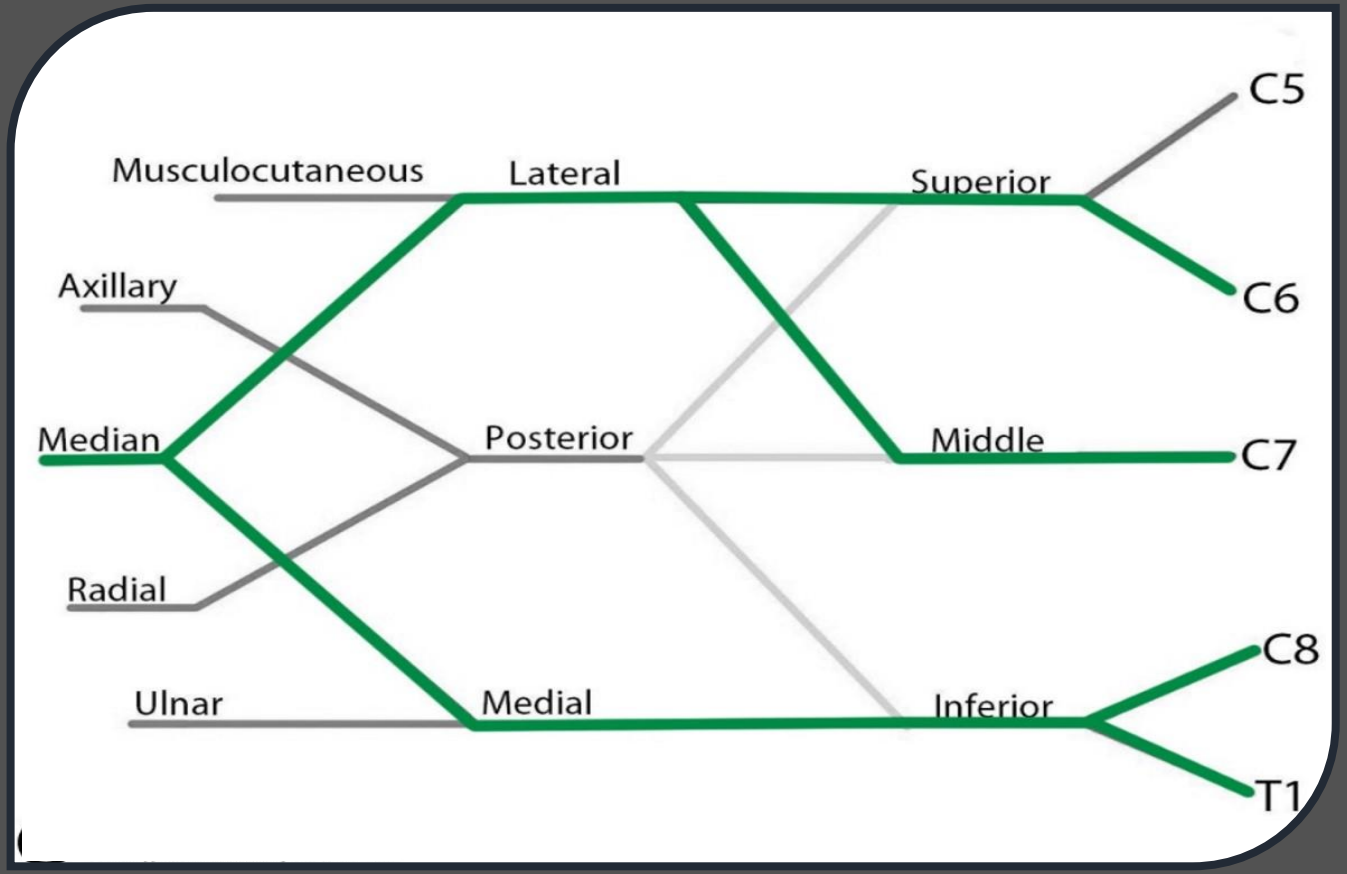
Brachial Plexus

- C5-T1 nerve roots

- Median nerve
- Ulnar nerve
- Radial nerve
- Axillary nerve
- Musculocutaneous nerve



Median Nerve



Median Nerve

Roots: C6 – T1

Motor Functions: Innervates most of the flexor muscles in the forearm, the thenar muscles, and the two lateral lumbrical muscles that move the index and middle fingers

Sensory Functions: Gives off the palmar cutaneous branch, which innervates the lateral part of the palm, and the digital cutaneous branch, which innervates the lateral three and a half fingers on the anterior (palmar) surface of the hand

Median Nerve



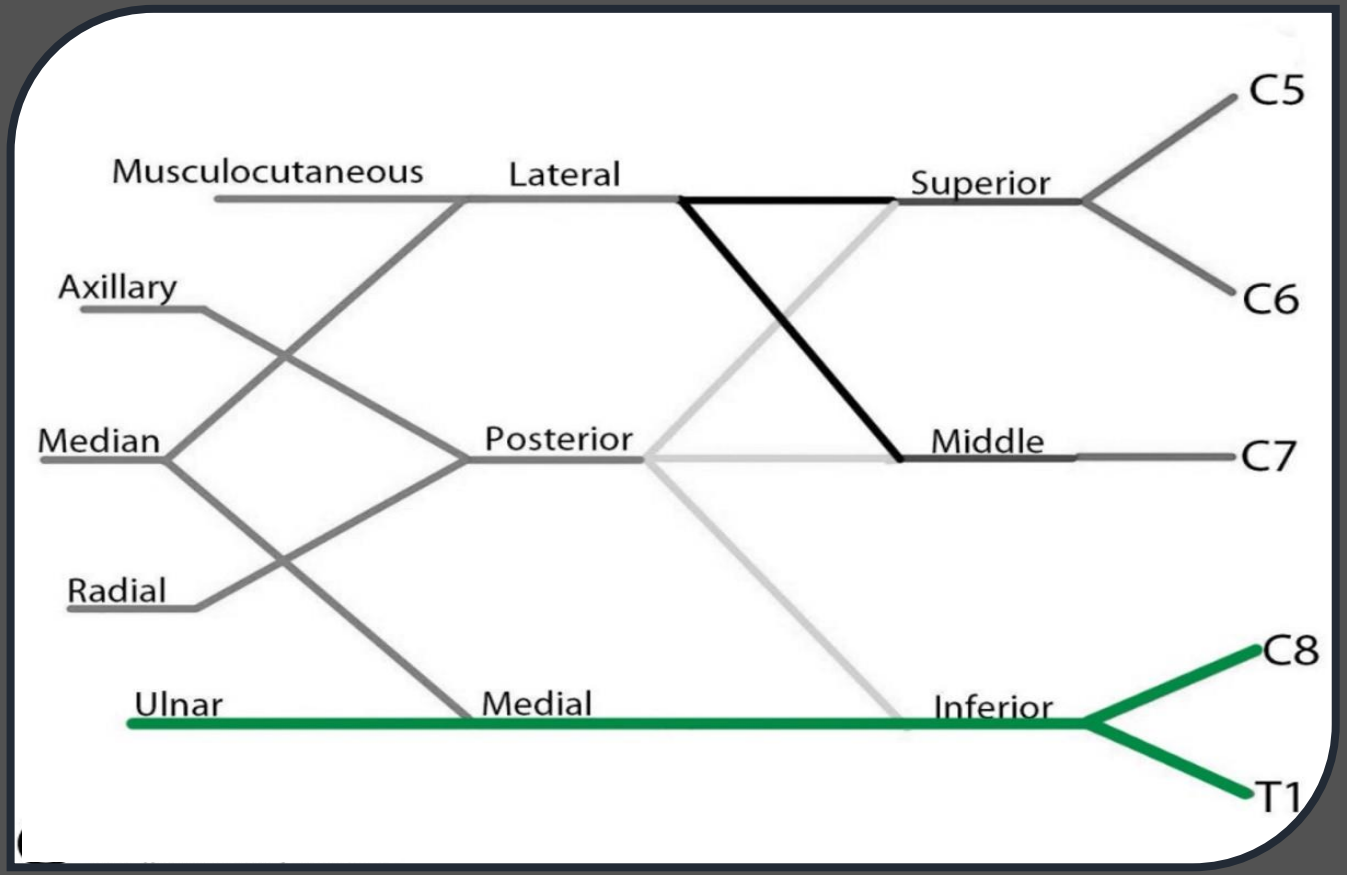
Common Median Nerve Diagnoses?

To diagnose Median Nerve pathology, look for a specific "territory" of symptoms and use provocative tests to reproduce the patient's pain:

- **Carpal Tunnel Syndrome** (wrist & elbow levels)
- **High median nerve palsy/ Hand of Benediction**
- **Pronator Syndrome**



Ulnar Nerve



Ulnar Nerve

Roots: C8 and T1

Motor Functions: Innervates the muscles of the hand, flexor carpi ulnaris and medial half of flexor digitorum profundus

Sensory Functions: Innervates the anterior and posterior surfaces of the medial one and half fingers, and associated palm area

Ulnar Nerve



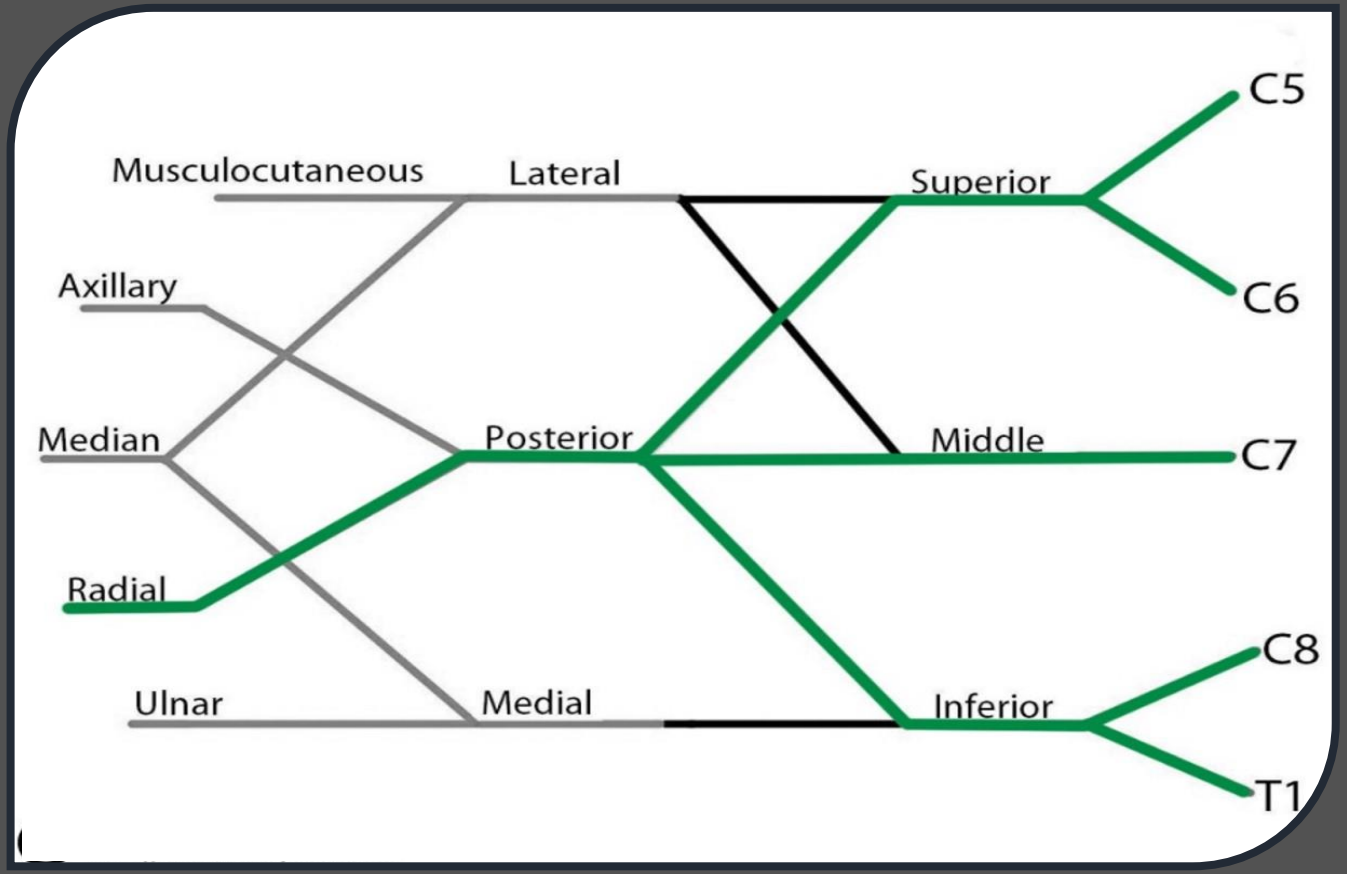
Common Ulnar Nerves Diagnoses?

- **Cubital Tunnel Syndrome (Elbow)**
- **Guyon’s Canal Syndrome (Wrist)**
- **Ulnar Claw Hand (Klumpke's Palsy)**

Clinical Pearl
Quick test: **Froment’s Sign**
& your Funny bone



Radial Nerve



Radial Nerve

Roots: C5 - T1

Motor Functions: innervates the triceps brachii, and the extensor muscles in the posterior compartment of the forearm

Sensory Functions: Innervates the posterior aspect of the arm and forearm, and the posterior, lateral aspect of the hand

Radial Nerve



Common Radial Nerve Diagnoses

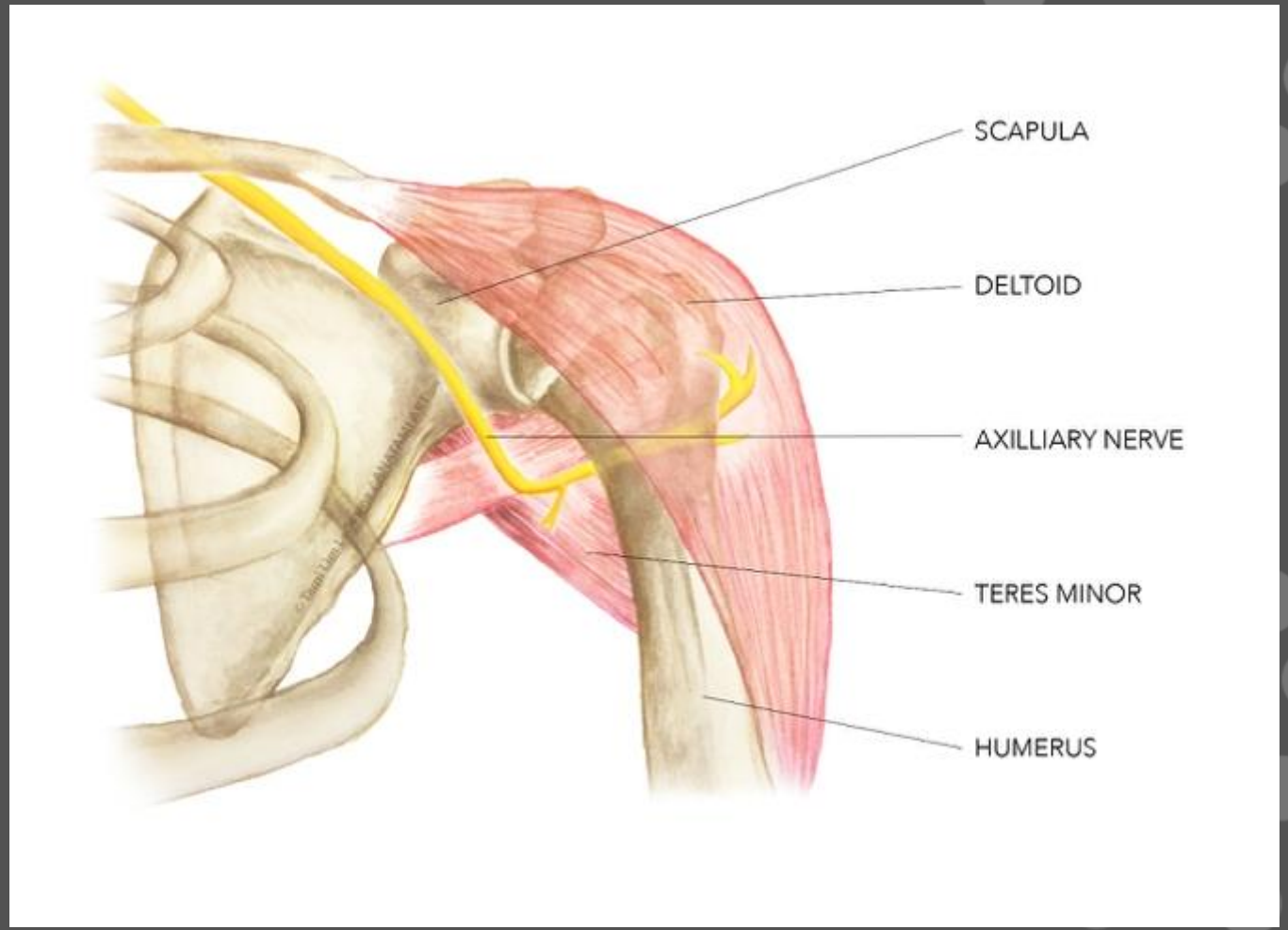
- **High Radial Nerve Compression (wrist drop)**
- **Radial Tunnel Syndrome**
- **Wartenberg's Syndrome (Cheiralgia Paresthetica**

Clinical Pearl
Quick test: **Hitchhikers test**



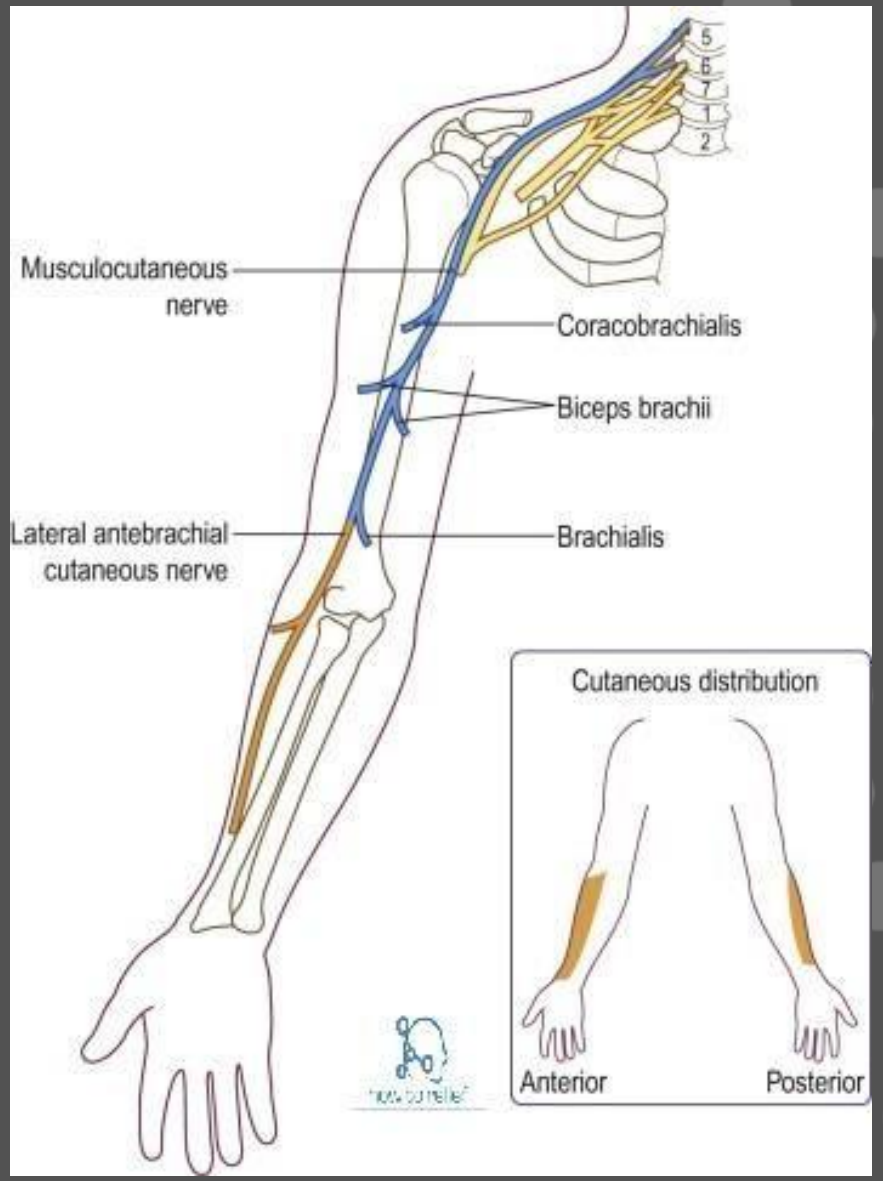
Axillary Nerve

- Roots C5-6
- Innervates Teres minor and Deltoid
- 3% to 6% of all brachial plexus pathologies
- Usually injured from compressive forces with anterior should dislocation



Musculocutaneous Nerve

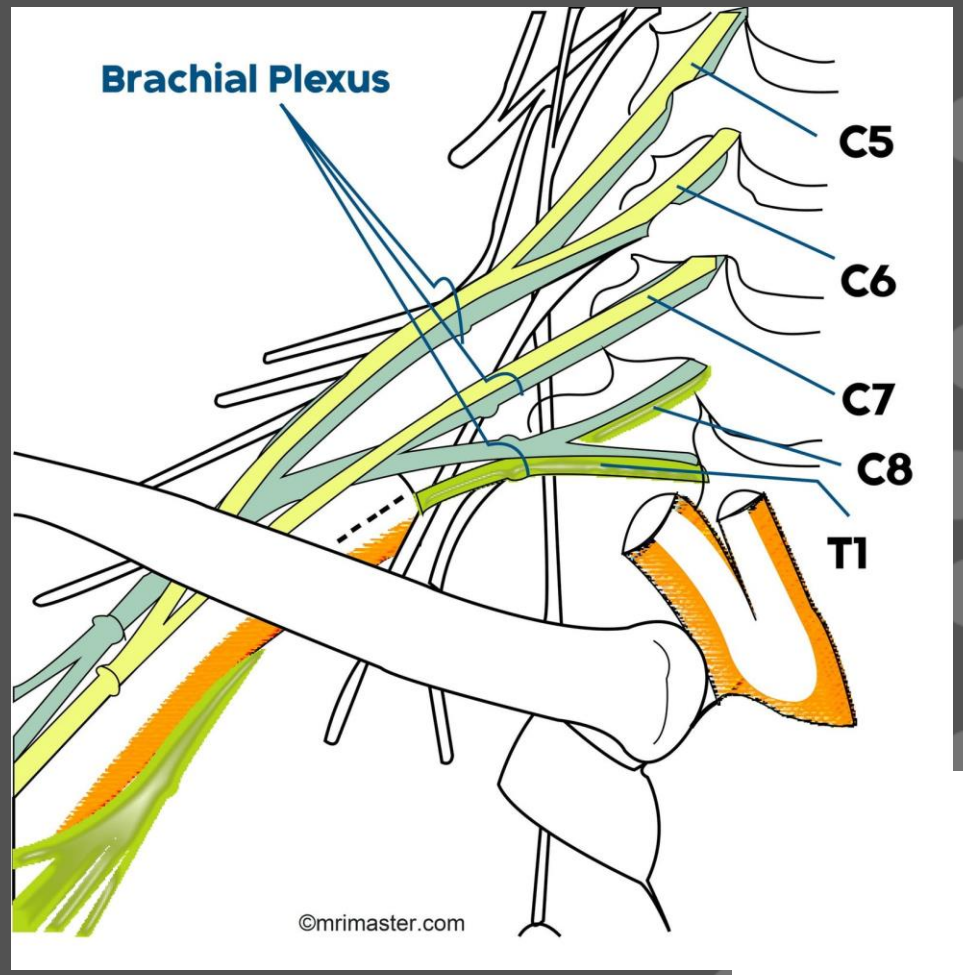
- Roots C5 – 7
- Innervates muscles at front of arm and skin of lateral forearm
- Innervates Biceps Brachii, Brachialis and Corocobrachialis

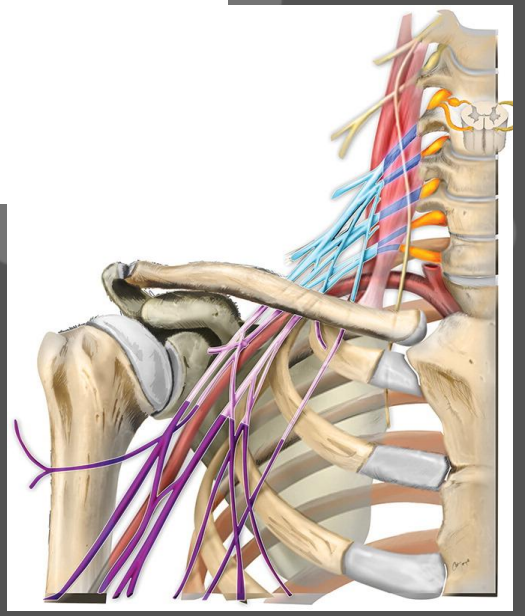


Brachial Plexopathy

- Dysfunction of the entire brachial plexus or specific portions of it (roots, trunks, divisions, cords, or branches)

- Palpate 1st rib





Summary

- Cervical vertebrae are smaller than Tx and Lx due to being less weightbearing
- C1-occipital, Atlanto-occipital joint (flexion/extension)
- C2-C1, Atlantoaxial joint (rotation)
- Cervical nerve roots
- Vertebrobasilar system passes through the cervical spine
  - Be aware of VBI

ANY QUESTIONS?

